

discipline-Level Analysis

2025-11-20

0. Order of Code Execution

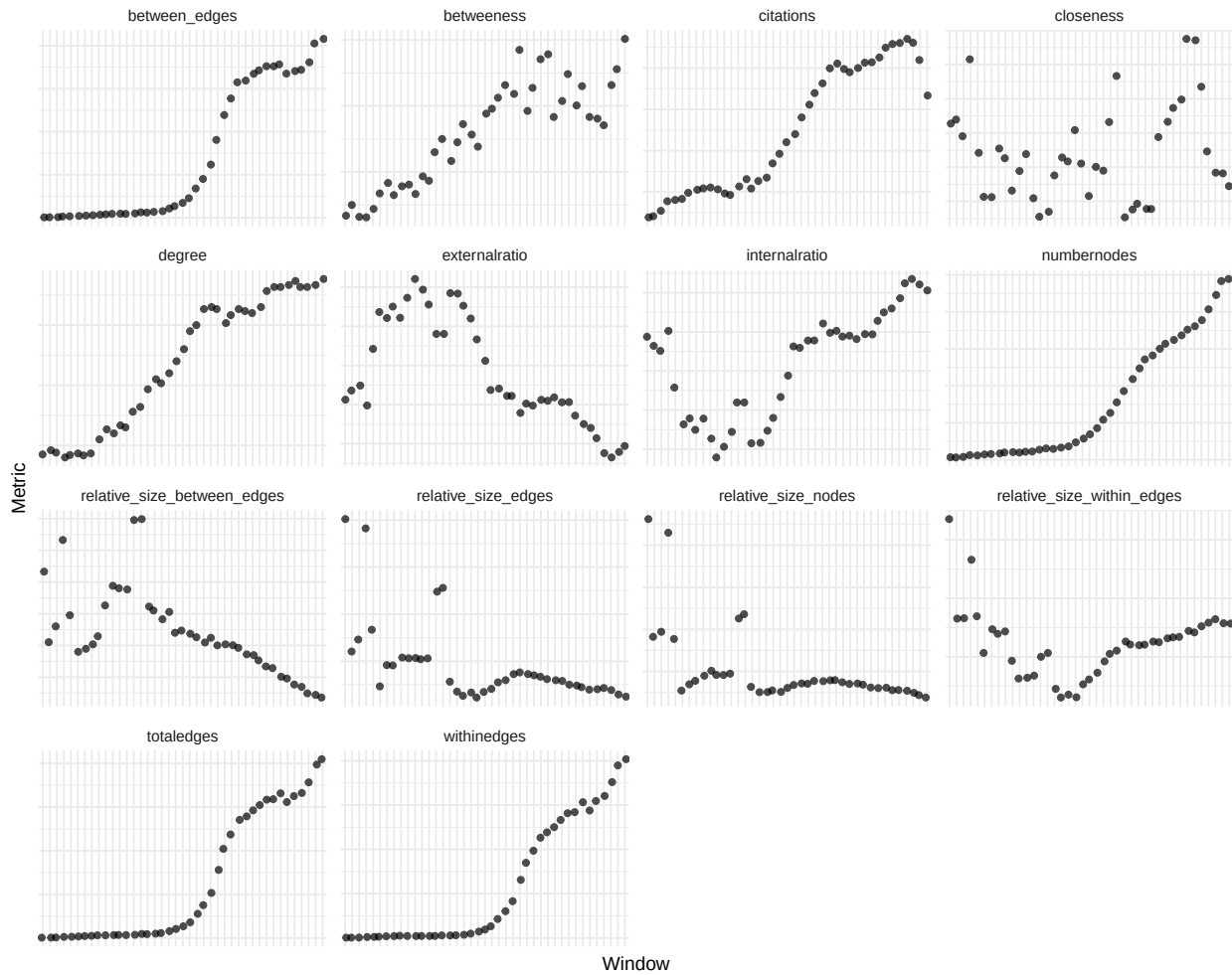
To generate the data used in this script the following scripts have to be executed:

“00_fetch_data_pairs.R” and “02_descriptives_and_cleaning.R” need to be executed to gather the OpenAlex data and clean it, respectively. This data is used to generate the temporal discipline-level networks based on bibliographic coupling in “03_generate_temporal_networks.R”. In “04_network_layout_field_supernodes.R” serves the main purpose to generate data for the interactive visualization of the networks (accessible via ...) and the summarization of the main meta information from the networks showing the evolution of disciplines (stored in folder “Evolution Fields”). “07_field_field_data.R” needs to be executed to obtain the discipline pair data and the results in Section 3, i.e. the discipline-discipline relationships. To study citation flows, first “00_scrape_references_fields.R” has to be executed in order to obtain journal data for the references of the articles under consideration. Subsequently “00_join_reference_disciplines.R” is executed to map discipline information to the journals similar to the methodology by Truc et al. (2023). After further computations in this script, the results in Section 4 are produced based on this data. To obtain the results in Section 5 abstract embeddings are generated with “generate_embeddings_v4.py”, which are then used as input to the BerTopic algorithm in “bertopic_simple_v4.py”, which infers the main topics within each discipline. The results are stored in “topics” folder. Furthermore, to analyze the discipline-level topical data “13_prep_donut_chart_data_js.R” might be executed which aggregates the article-level topic data to a discipline level.

1. Aggregate Metrics

The Metrics used here are formally defined on a per-discipline basis below. The aggregate metrics are computed similarly, but over all disciplines instead of per discipline.

1.1 Evolution Aggregate of Metrics



1.2 Number of Nodes and Edges over Time

Window	Number of Nodes	Number of Edges
1980-1984	63	208
1981-1985	76	279
1982-1986	82	345
1983-1987	117	562
1984-1988	147	792
1985-1989	183	935
1986-1990	201	1034
1987-1991	229	1219
1988-1992	260	1545
1989-1993	276	1554
1990-1994	268	1596
1991-1995	297	1852
1992-1996	287	1729
1993-1997	288	1525
1994-1998	320	1922
1995-1999	364	2177
1996-2000	396	2180
1997-2001	416	2305
1998-2002	472	2762
1999-2003	525	3493
2000-2004	564	4020
2001-2005	636	5238
2002-2006	735	7773
2003-2007	865	10487
2004-2008	994	12811
2005-2009	1174	19481
2006-2010	1333	24829
2007-2011	1530	30684
2008-2012	1741	37964
2009-2013	1865	41095
2010-2014	2075	46937
2011-2015	2289	53723
2012-2016	2491	57817
2013-2017	2712	57800
2014-2018	3014	66179
2015-2019	3319	65021
2016-2020	3661	71720
2017-2021	4110	76929
2018-2022	4686	87002
2019-2023	5155	100334
2020-2024	5340	104634

2. Evolution of disciplines

discipline-Level Metrics are derived from networks where articles are nodes and edges are based on the strength of bibliographic coupling between articles. The nodes, edges and their information is then aggregated per discipline, here called discipline.

2. Relative Number of Nodes and Number of Citations

- Relative number of Nodes:

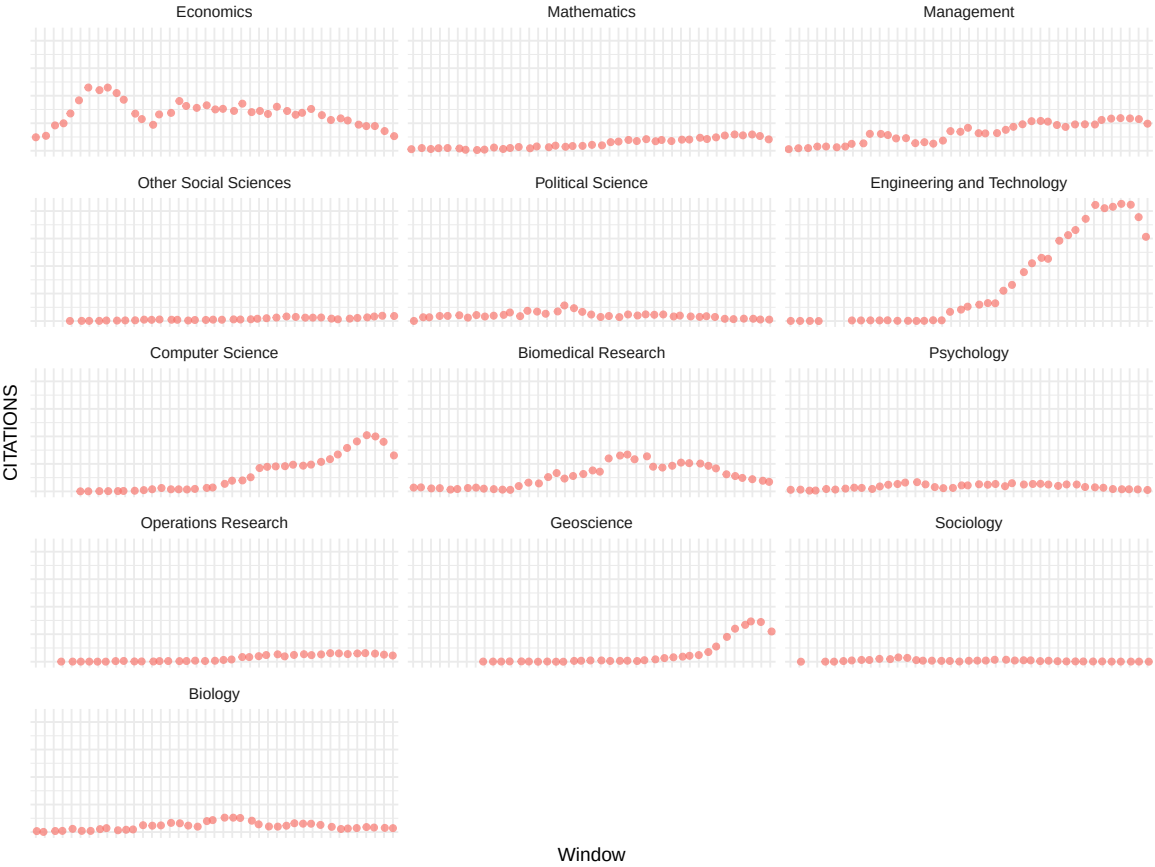
$$RelNodes_{t,f} = \frac{N(V)_{t,f}}{\sum_f^F N(V)_{t,f}} \quad (1)$$

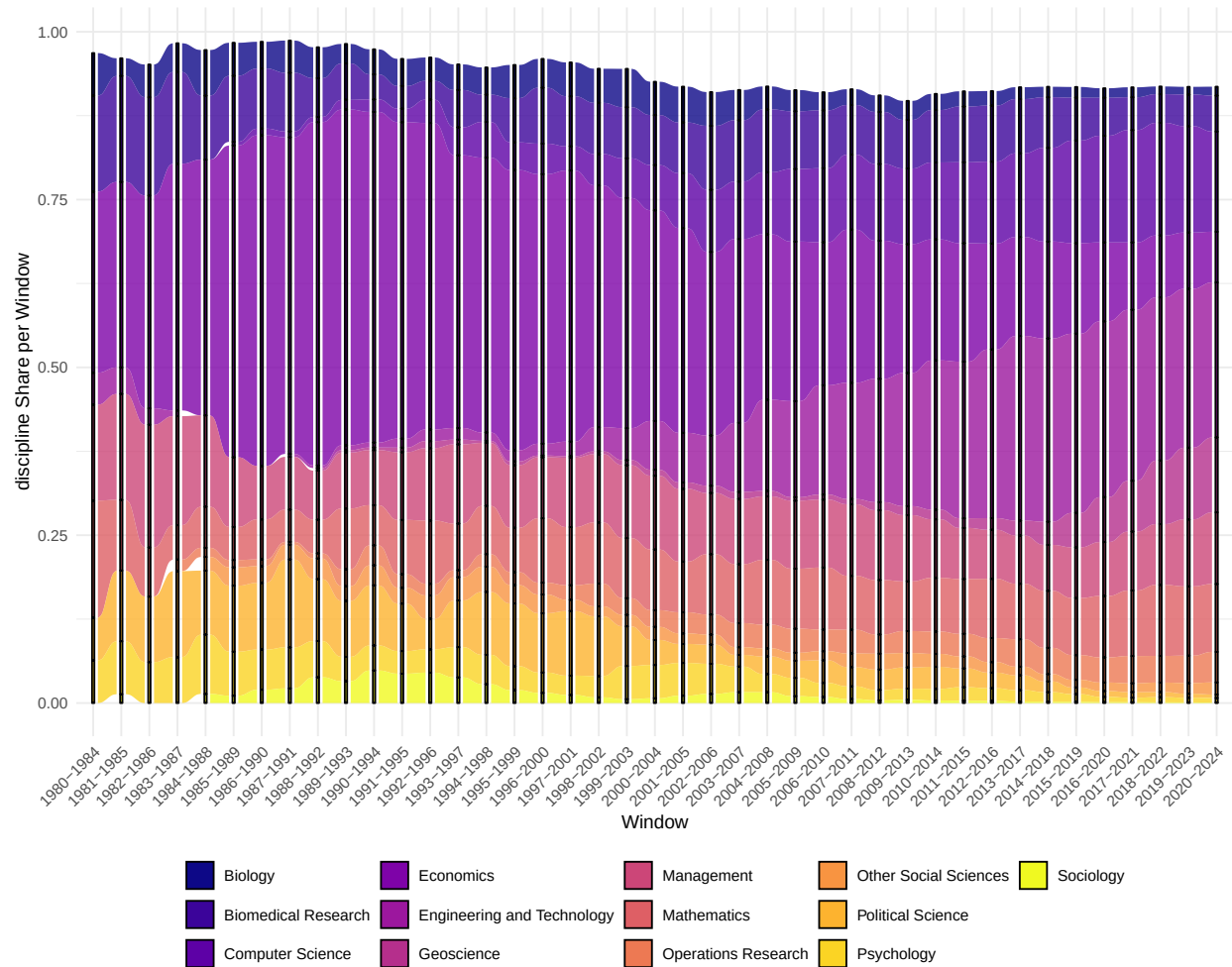
where:

- $RelNodes_{t,f}$ measures the relative number of nodes for discipline f in time window t across all article nodes
- $N(V)_{t,f}$ measures the total number of nodes for discipline f in time window t across all article nodes. It is therefore defined as $N(V)_{t,f} = \sum_{v_i} 1_{discipline(v_i)=f}$, where $1_{discipline(v_i)=f}$ is an indicator function, taking the value one if article node v_i belongs to discipline f .
- The number of citations per discipline and time window is measured as the sum of citations across all articles of a discipline in a give time window, respectively.



Evolution of Citations per discipline





2.2 Relative Number of Edges

- Relative Number of Total Edges:

$$RelEdges_{t,f} = \frac{N(E)_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (2)$$

where:

- $RelEdges_{t,f}$ measures the relative number of edges incident on nodes within discipline f in time window t
- $N(E)_{t,f}$ measures the total number of edges incident on nodes within discipline f in time window t . It is therefore defined as $N(E)_{t,f} = \sum_{e=(v_i,v_j)} 1_{discipline(v_i)=f \vee discipline(v_j)=f}$, where $1_{discipline(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to discipline f
- Relative Number of Within-discipline Edges:

$$RelEdgesWithin_{t,f} = \frac{N(E_{within})_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (3)$$

where:

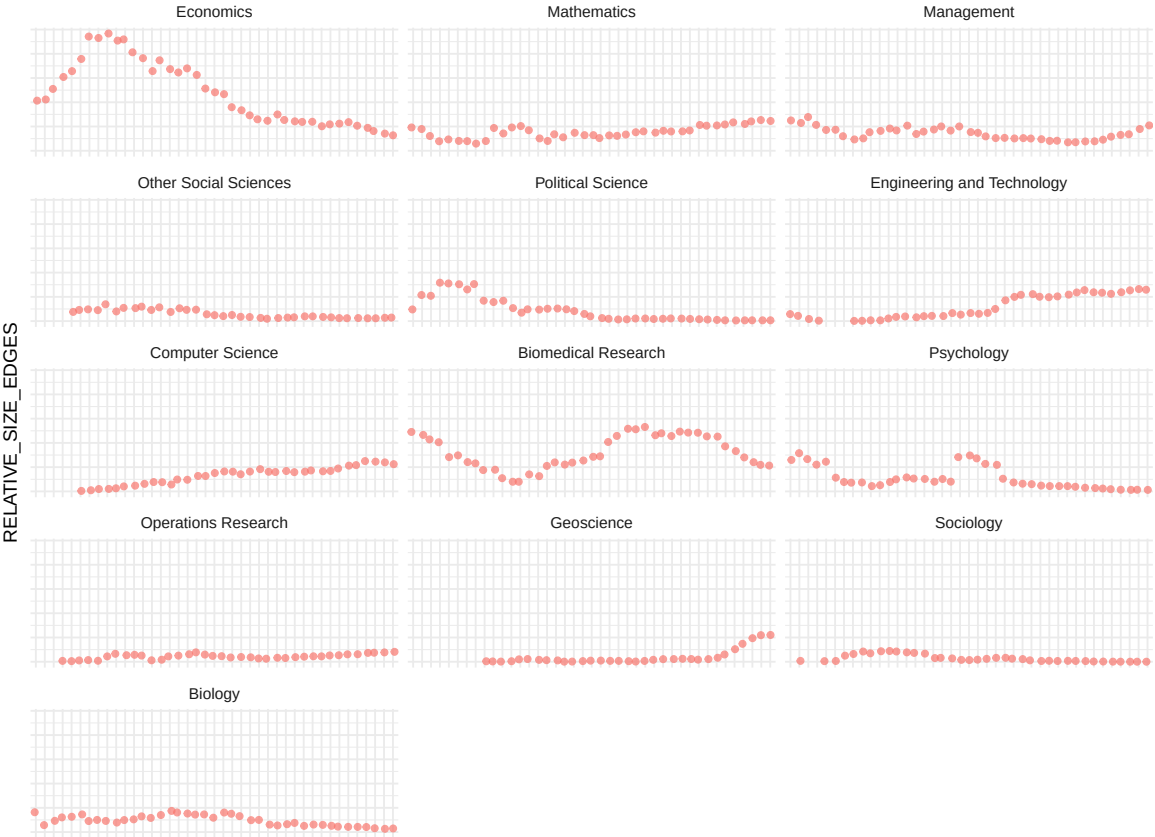
- $RelEdgesWithin_{t,f}$ measures the relative number of edges between article nodes of the same discipline for discipline f in time window t
- $N(E_{within})_{t,f}$ measures the total number of edges between article nodes of the same discipline for discipline f in time window t . It is therefore defined as $N(E_{within})_{t,f} = \sum_{e=(v_i,v_j)} 1_{discipline\{(v_i)=f\}} 1_{\{discipline(v_j)=f\}}$, where $1_{discipline(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to discipline f .
- Relative Number of Between-discipline Edges:

$$RelEdgesBetween_{t,f} = \frac{N(E_{Between})_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (4)$$

where:

- $RelEdgesBetween_{t,f}$ measures the relative number of edges between article nodes of different disciplines for discipline f in time window t
- $N(E_{Between})_{t,f}$ measures the total number of edges between article nodes of different disciplines for discipline f in time window t . It's defined as $N(E_{Between})_{t,f} = \sum_{e=(v_i,v_j)} (1_{discipline\{(v_i)=f\}} 1_{discipline\{(v_j) \neq f\}} + 1_{discipline\{(v_i) \neq f\}} 1_{discipline\{(v_j)=f\}})$

Evolution of Relative Number of Edges per discipline



Window

Evolution of Relative Number of Edges within/between disciplines



2.3 Internal and External Edge Ratio

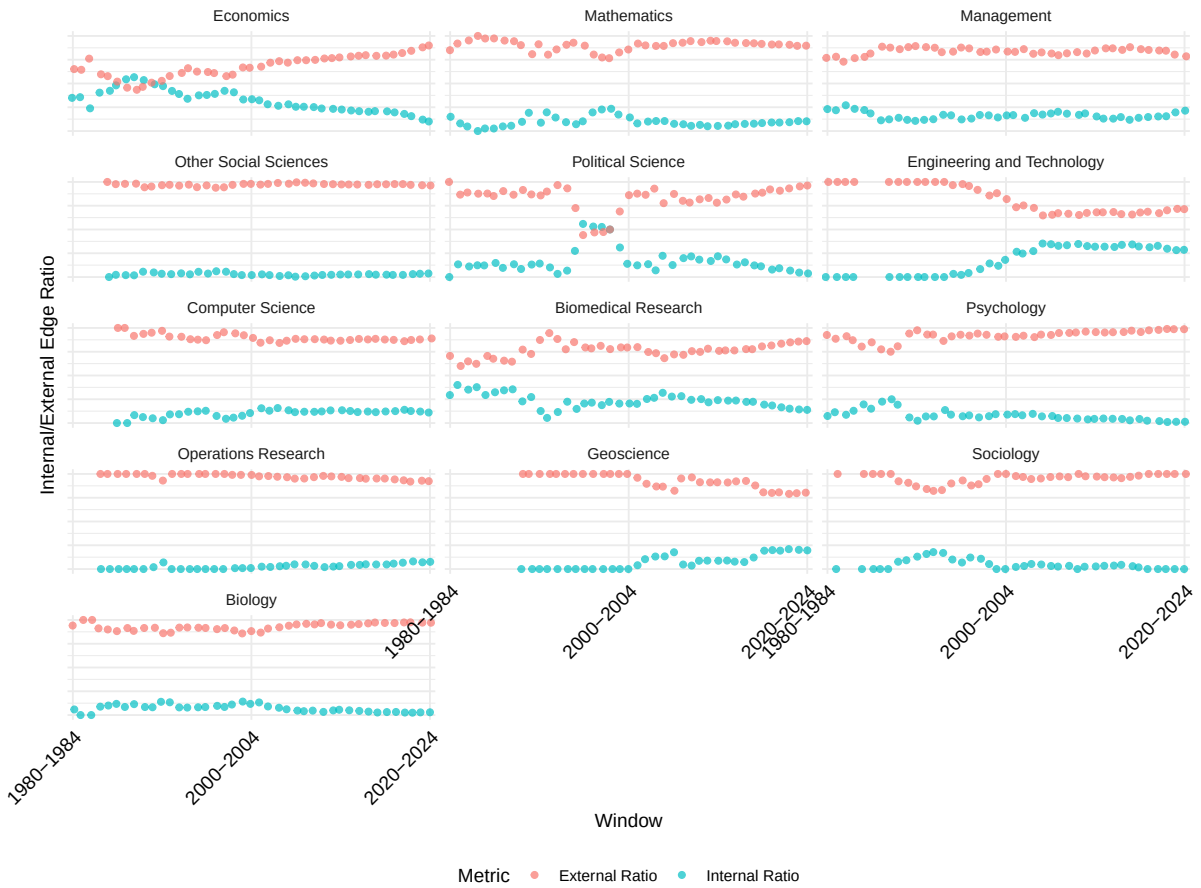
- Internal Edge Ratio:

$$IR_{t,f} = \frac{N(E_{within})_{t,f}}{N(E)_{t,f}} \quad (5)$$

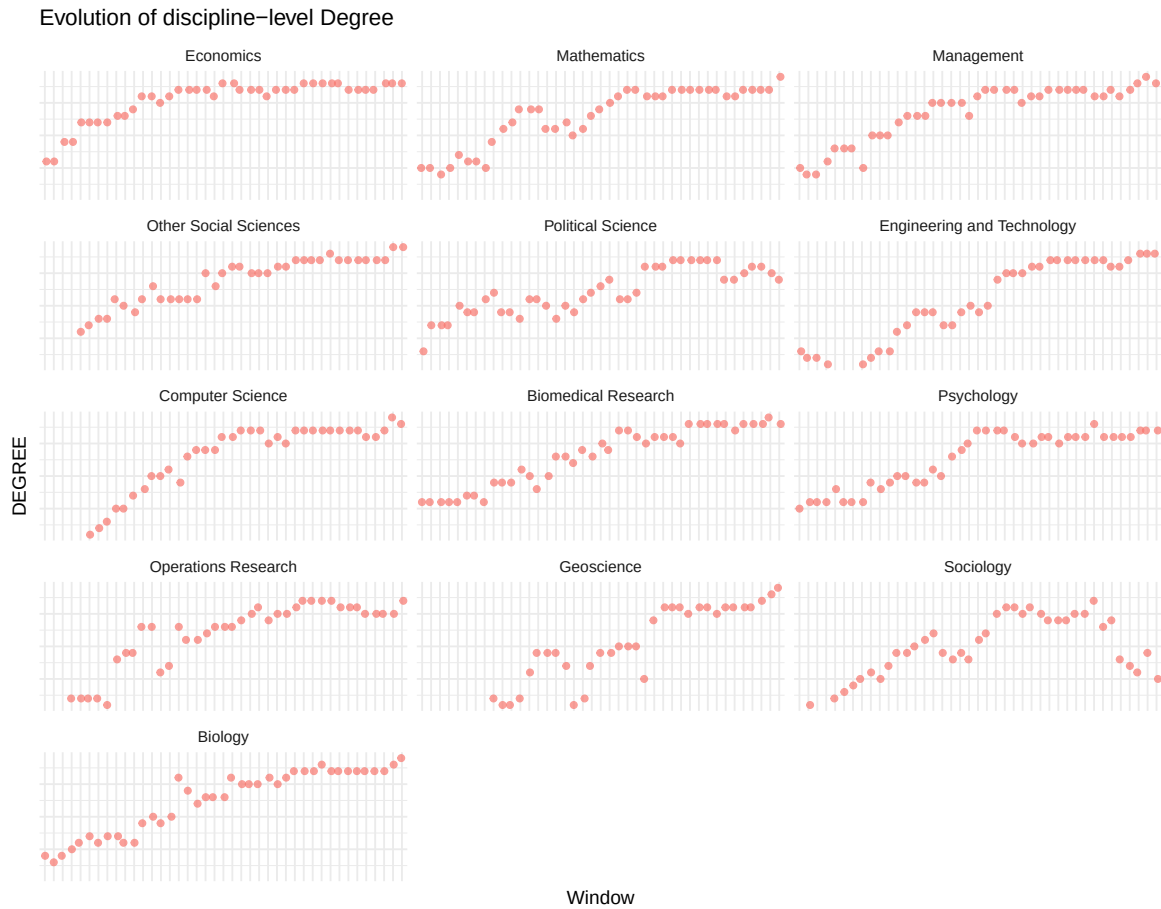
- External Edge Ratio:

$$ER_{t,f} = \frac{N(E_{between})_{t,f}}{N(E)_{t,f}} \quad (6)$$

with $N(E_{within})_{t,f}$, $N(E_{between})_{t,f}$ and $N(E)_{t,f}$ defined as before. The internal and external ratios therefore measure the ratio of edges within (between) disciplines relative to the total number of edges of a discipline. Edges are derived from article-level nodes.



2.4 Network Centrality Metrics (Degree, Closeness and Betweenness)



Evolution of discipline-level Closeness

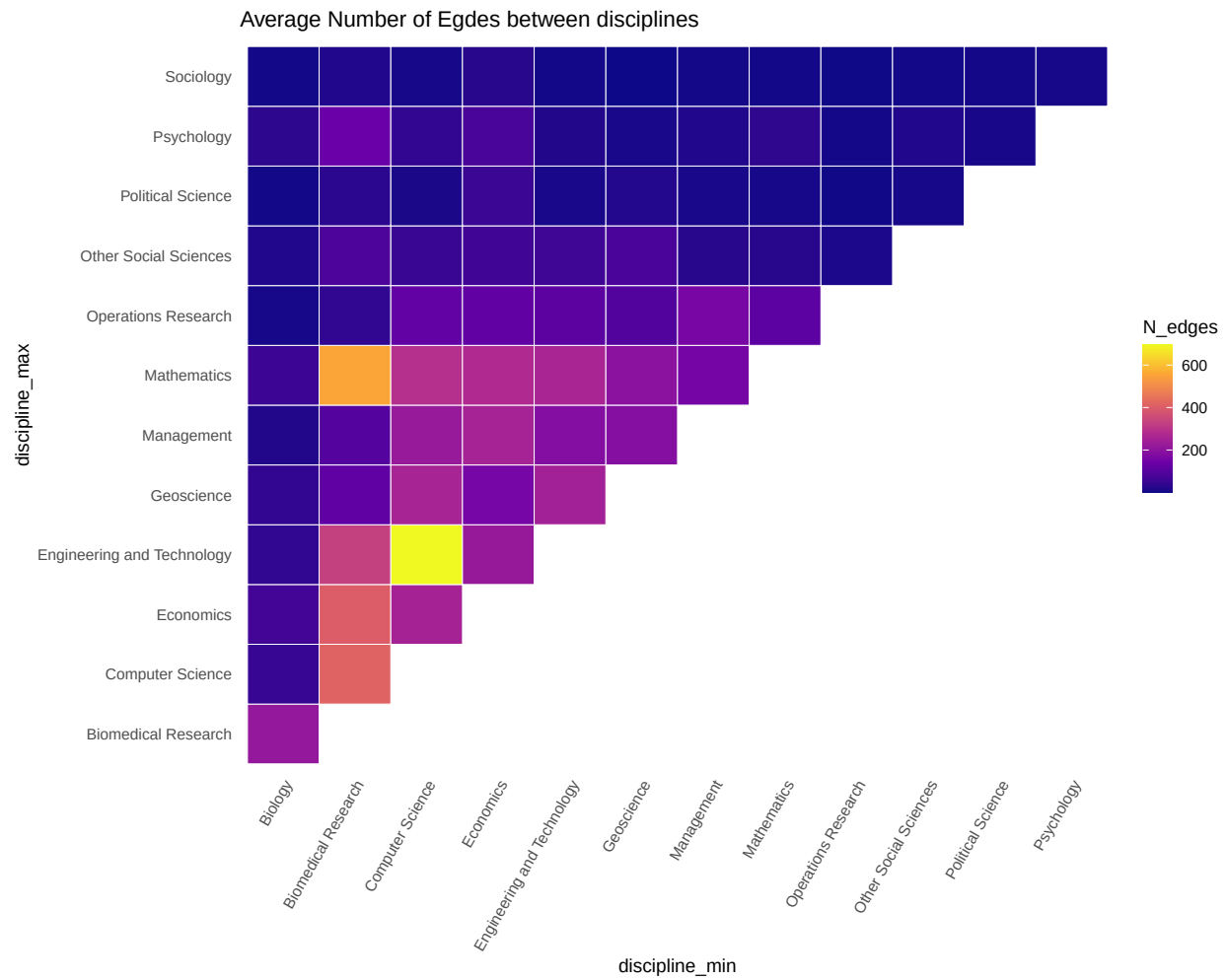


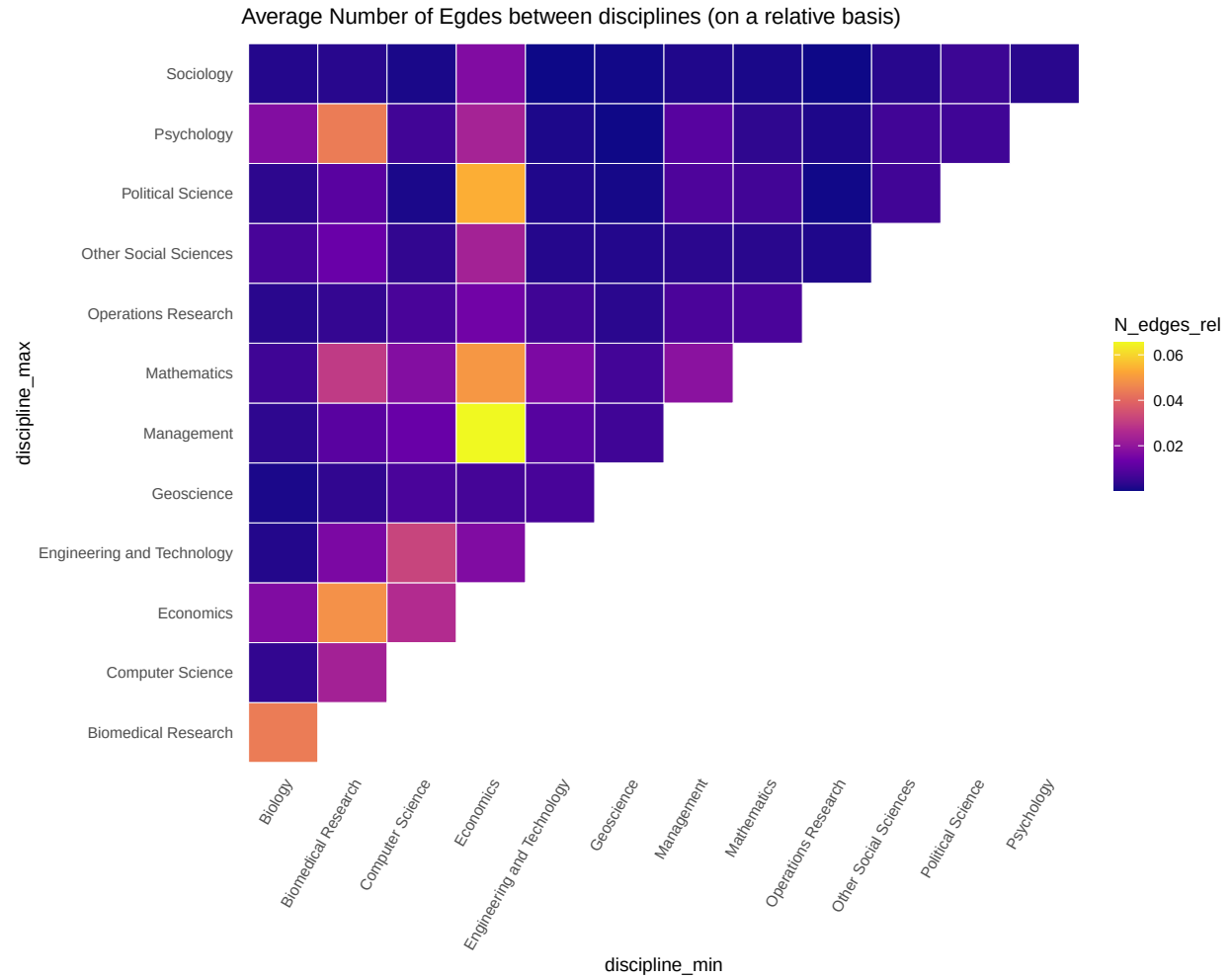
Evolution of discipline-level Betweenness



3. discipline-discipline Connectivity

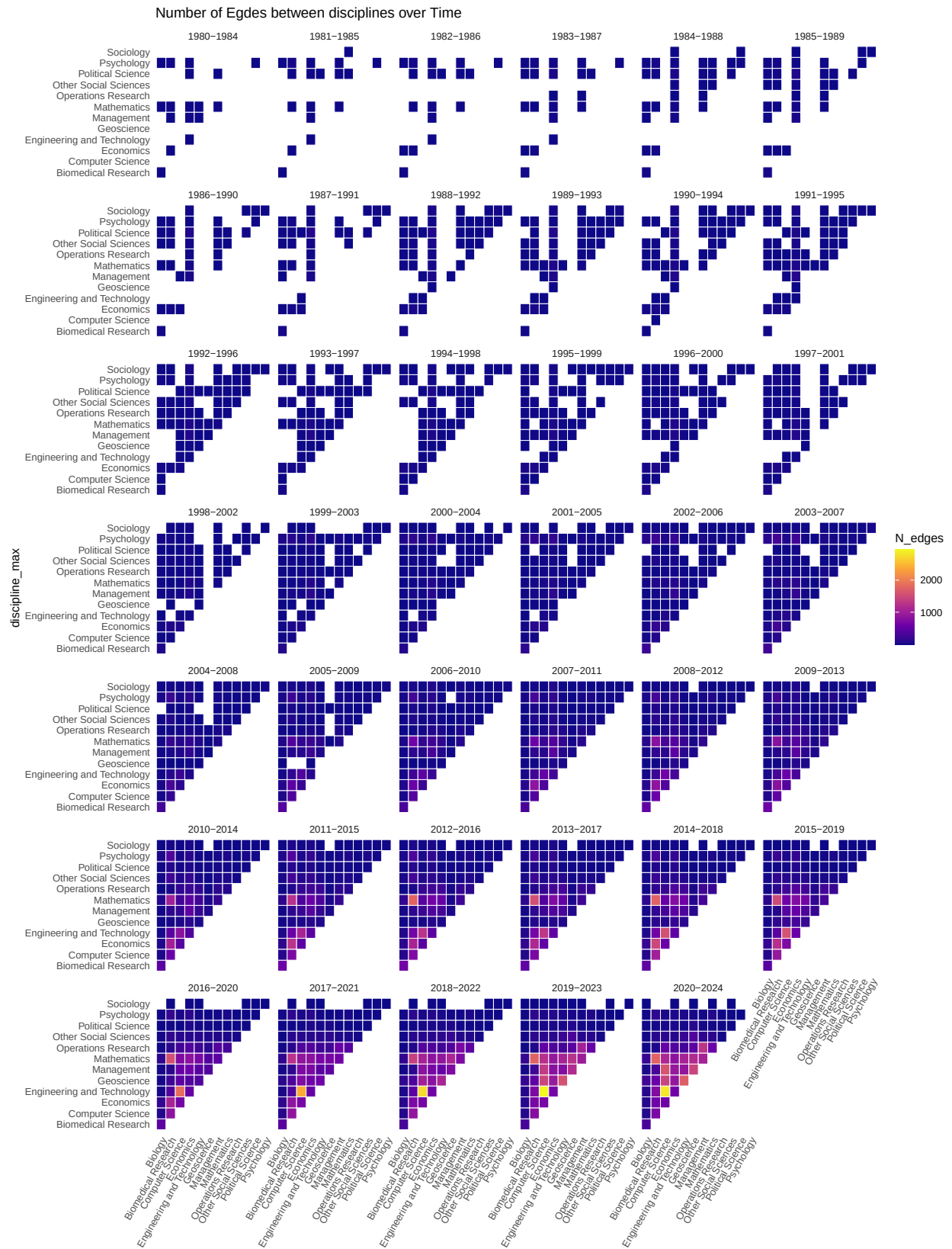
3.1 Average Connectivity



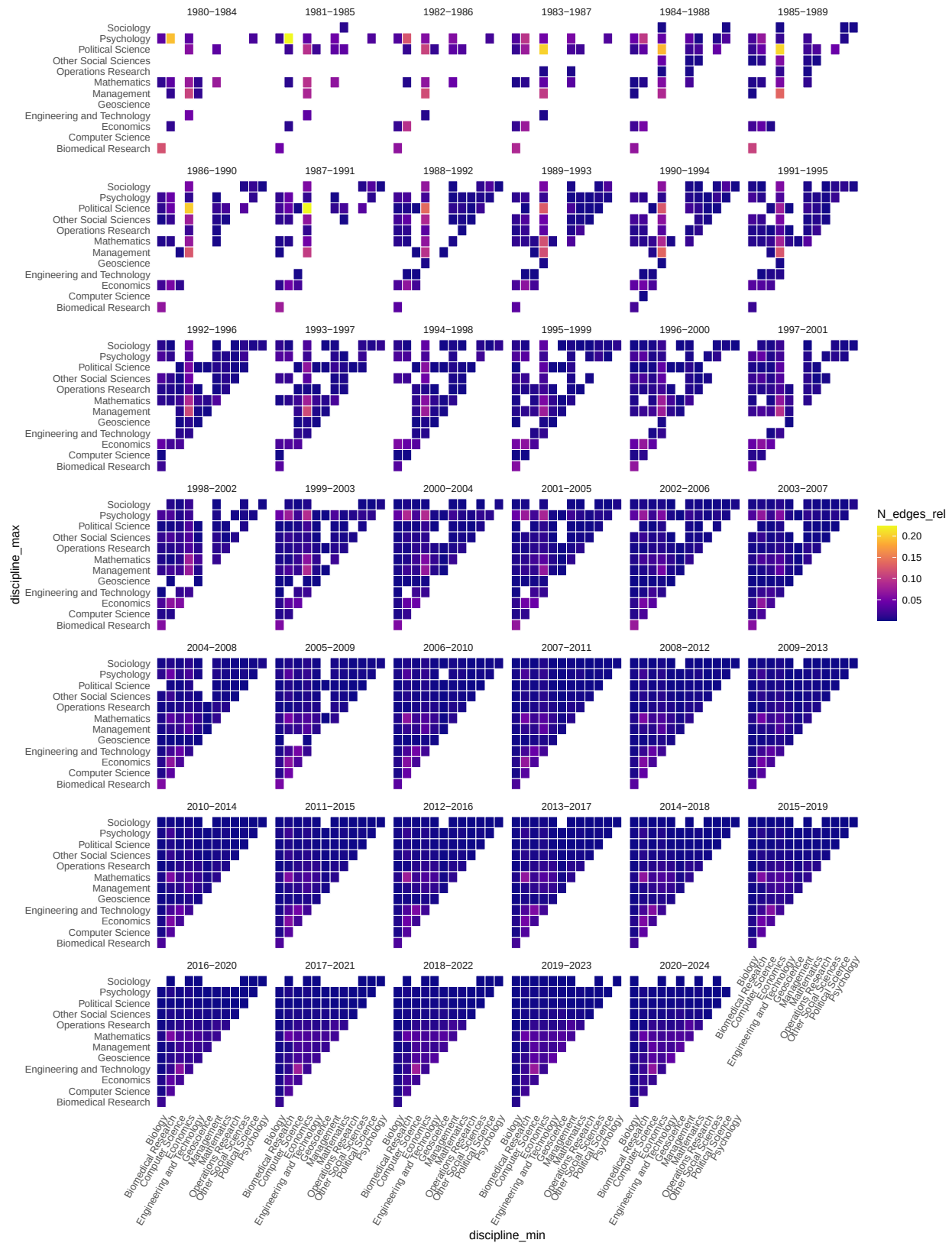


3.2 Connectivity over time

3.2.1 Heatmaps

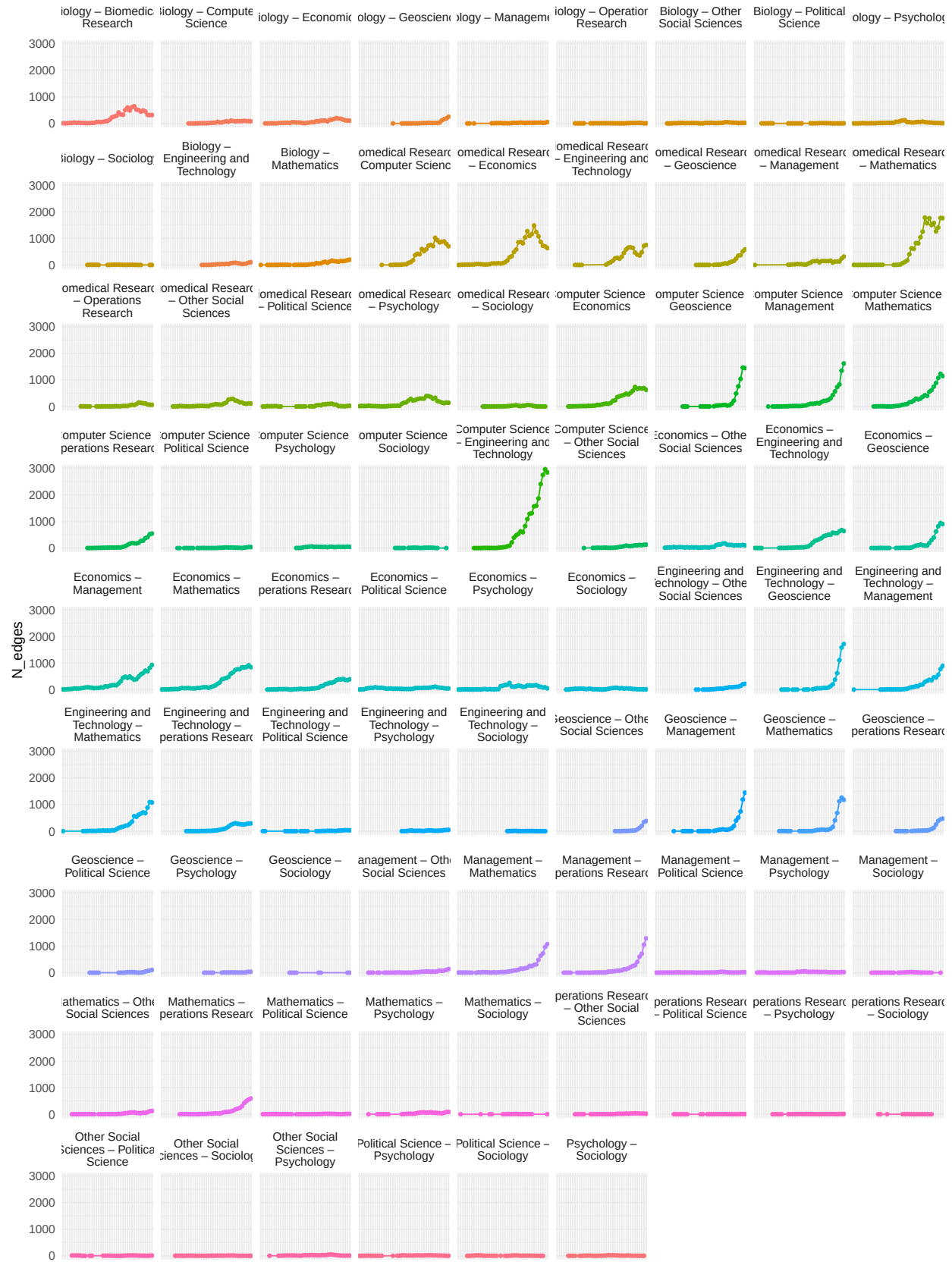


Number of Edges between disciplines over Time (on a relative basis)

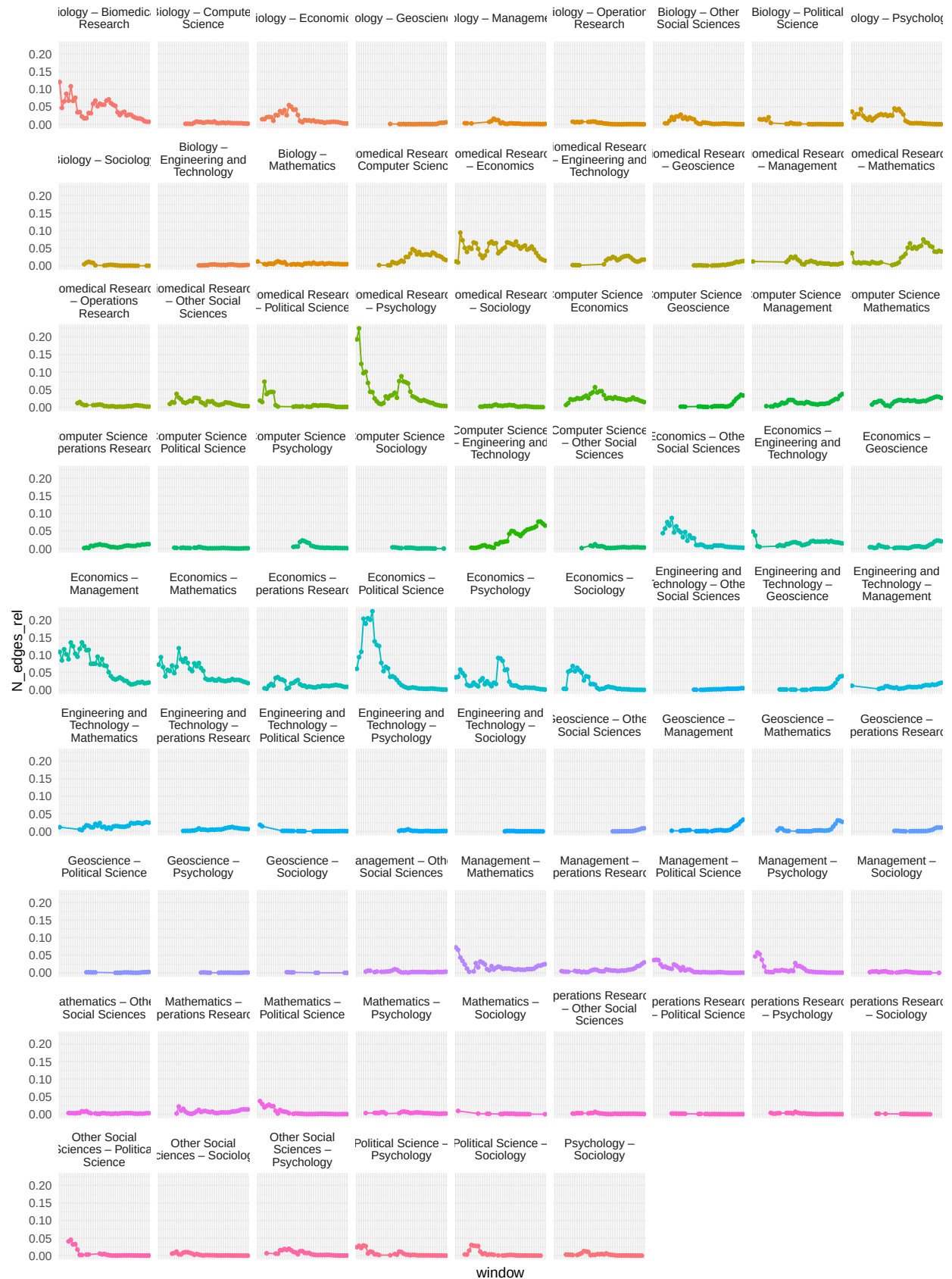


3.2.2 Time Series

Number of Edges between disciplines over Time



Number of Edges between disciplines over Time (on a relative basis)



4. Citation Flows

4.1 Internal Coverage

Table 1: Number of References with Meta Data

discipline	Total Number of References	Relative Number of References (in %)
Arts	115	0.02
Biology	15414	2.22
Biomedical Research	36964	5.31
Chemistry	561	0.08
Computer Science	40252	5.78
Economics	87319	12.55
Engineering and Technology	79876	11.48
Geoscience	22792	3.28
Health	1706	0.25
Humanities	2440	0.35
Law	1408	0.20
Management	61957	8.90
Mathematics	28286	4.06
Medicine	7368	1.06
Operations Research	10914	1.57
Other Professional Fields	9199	1.32
Other Social Sciences	8888	1.28
Physics	19394	2.79
Political Science	11242	1.62
Psychology	13148	1.89
Sociology	3359	0.48
NA	233275	33.52

4.2 Outside/Inside Citations

4.2.1 Per Year



Outside/Inside Reference per Year



4.2.2 Per Window (non-overlapping)



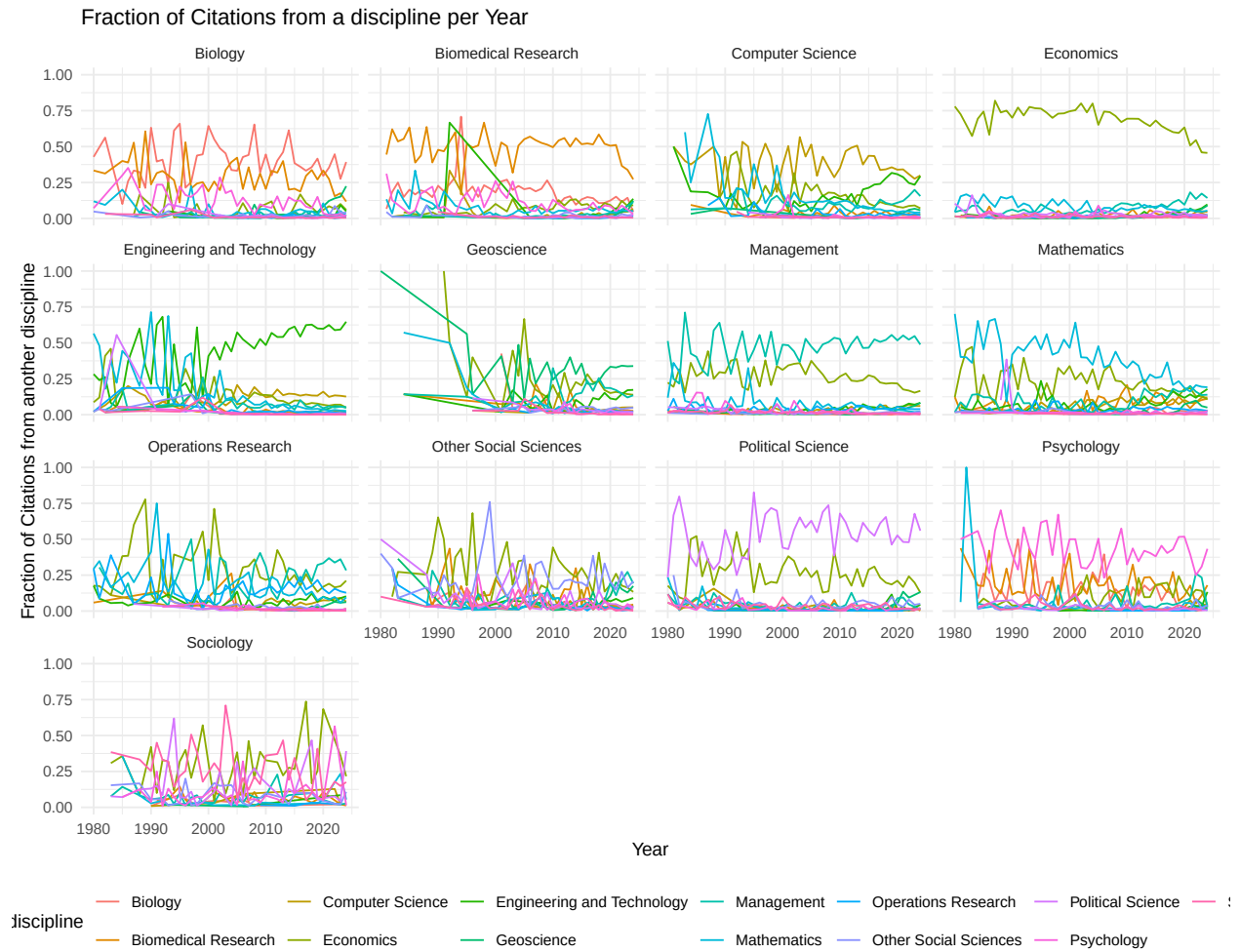
Outside/Inside Reference per Window



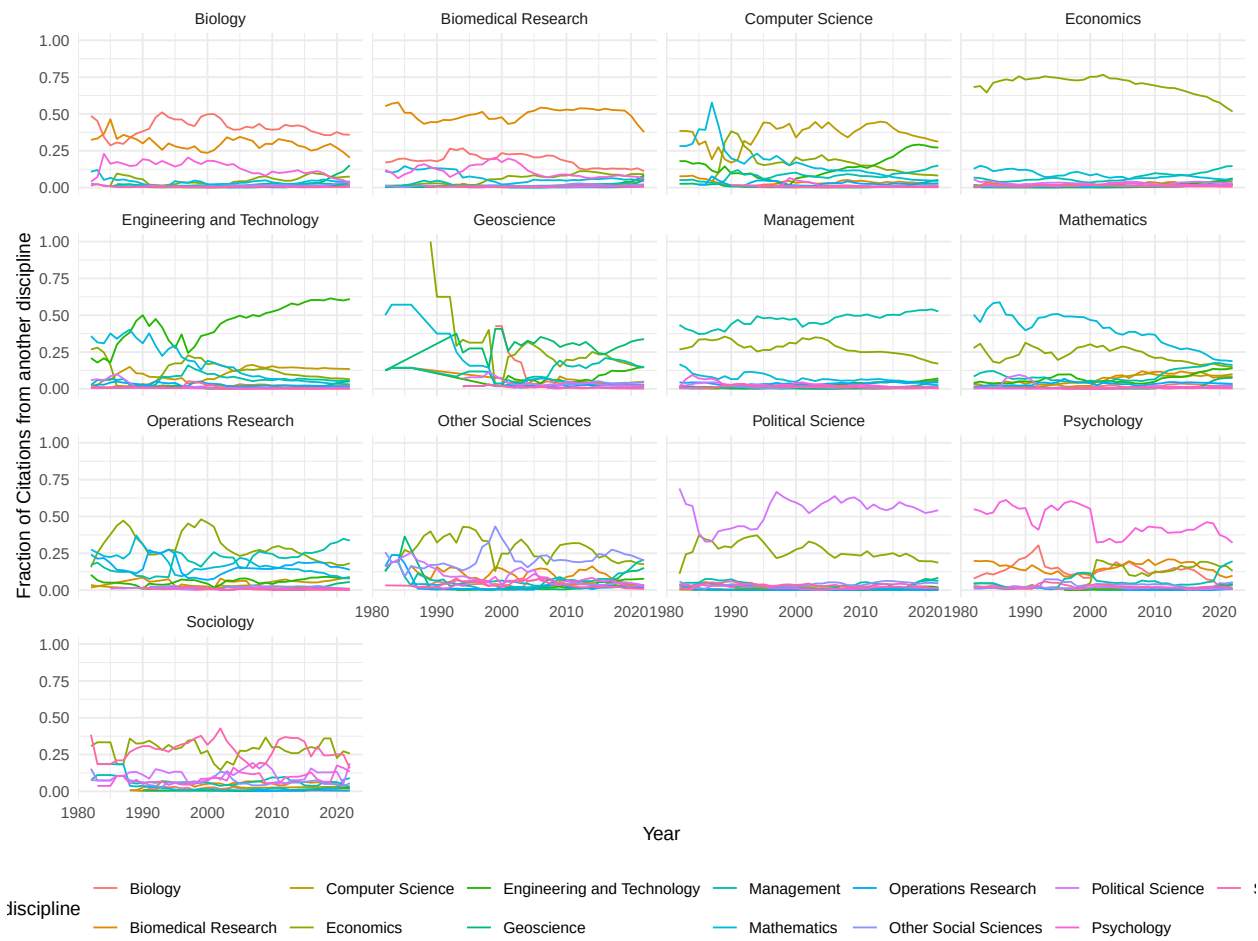
4.2.3 Per Window (Overlapping)



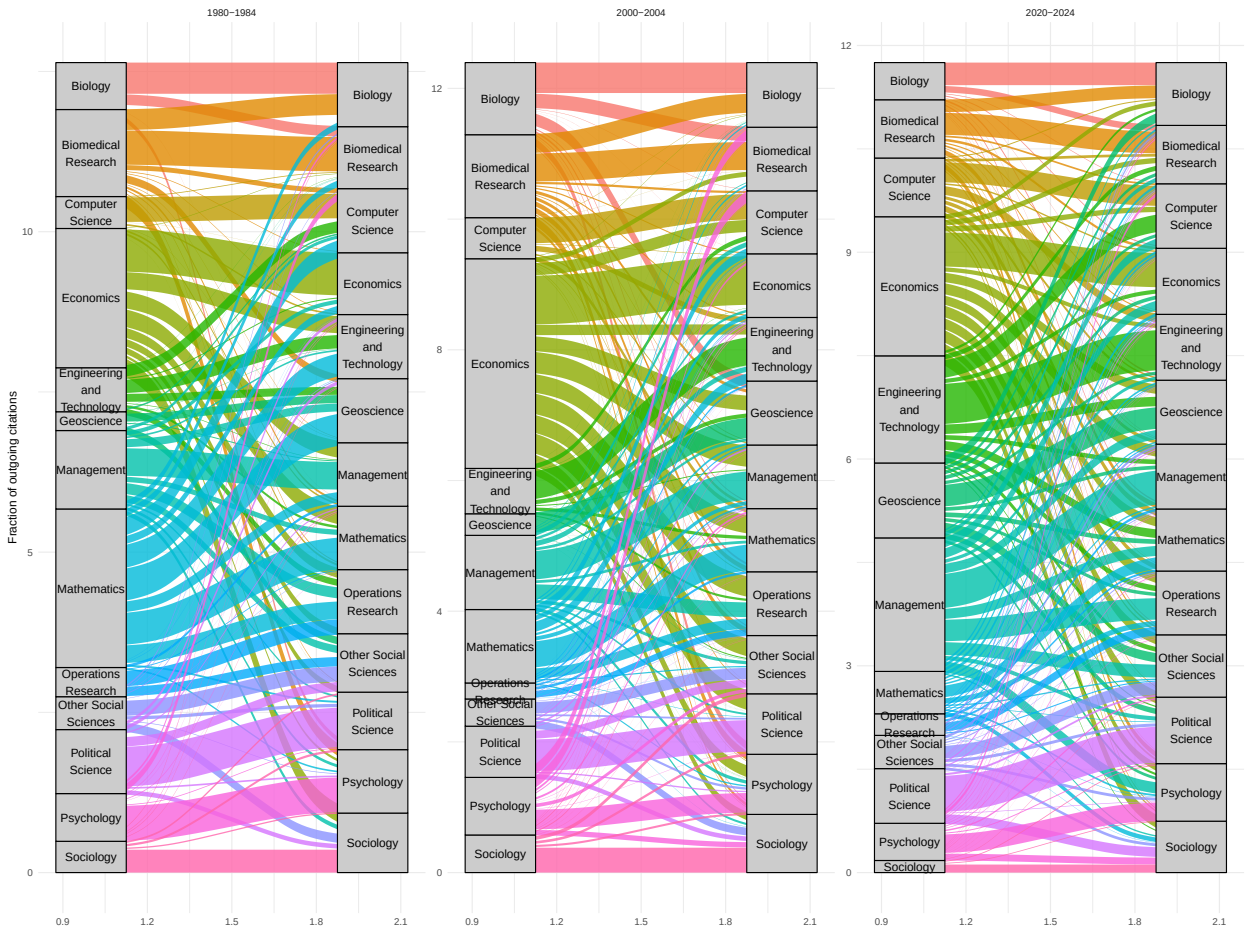
4.3 Citation Flows over Time



Fraction of Citations from a discipline per 5-year Rolling Windows



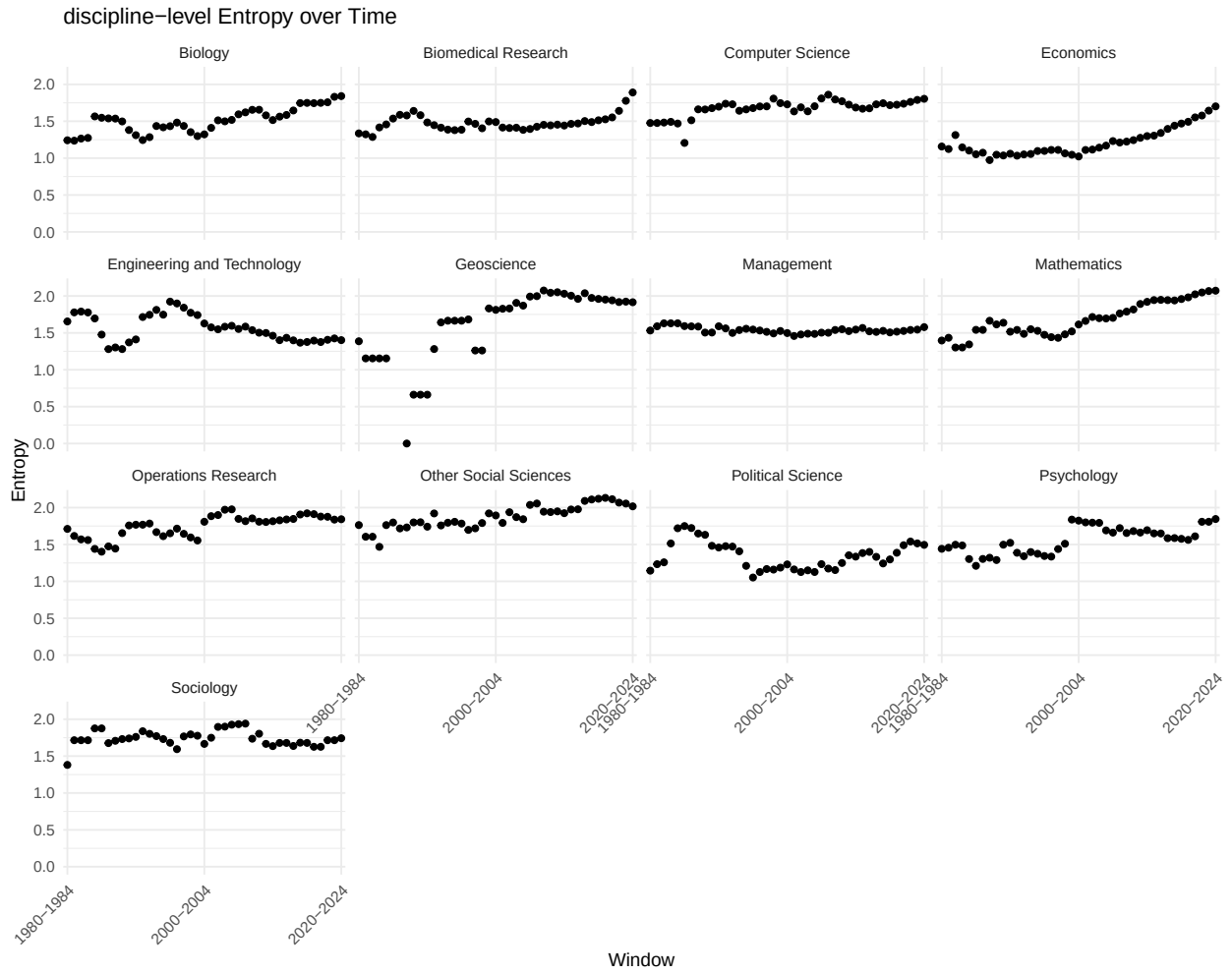
4.4 Alluvial Graph of Flows



4.5 Citation Flow Dispersion

Table 2: Citation Flow Entropy across disciplines

discipline	Flow Entropy
Biology	61.87831
Biomedical Research	60.64829
Computer Science	68.60920
Economics	49.71776
Engineering and Technology	64.05751
Geoscience	62.95965
Management	63.04808
Mathematics	68.93830
Operations Research	71.56687
Other Social Sciences	76.83326
Political Science	55.20307
Psychology	63.95801
Sociology	71.27874



5. Topical Analysis of disciplines

5.1 Topical Overview

Table 3: Overview of Topics within disciplines.

discipline	Topic No.	Keywords	No. Articles
Biology	0	prey, males, male, food, mating, female, game, model, females, behavior	53
Biology	1	government, marine, game, management, enterprises, development, study, governance, carbon, food	46
Biology	2	cooperation, game, social, games, trust, selection, evolutionary, defectors, strategies, players	22
Biology	3	plant, plants, height, growth, water, root, hydraulic, leaf, soil, trees	16
Biology	4	assessment, contest, fighting, size, strategies, contests, models, animal, escalate, opponents	10
Biology	5	species, mml, evolutionary, population, dynamics, bifurcation, structured, ifd, equations, darwinian	10
Biology	6	lewontin, citation, information, et, al, world, time, genetics, state, articlegoogle	7
Biology	7	virulence, cancer, evolution, host, cells, treatment, resistance, pathogen, evolutionary, dynamics	7
Biology	8	disease, vaccination, vaccine, equilibria, model, distancing, population, game, level, mixing	6
Biomedical Research	0	cooperation, game, social, group, evolution, theory, individuals, model, helping, evolutionary	118
Biomedical Research	1	government, game, evolutionary, strategy, carbon, chain, enterprises, model, supply, low	77
Biomedical Research	2	evolutionary, population, game, dynamics, host, model, species, strategies, selection, strategy	75
Biomedical Research	3	mml, games, game, reasoning, mn, strategies, competitive, strategic, systems, theory	26
Biomedical Research	4	vaccination, epidemic, disease, control, public, game, model, vaccine, individual, individuals	25
Biomedical Research	5	sperm, male, males, female, sex, care, reproductive, mating, size, competition	21
Biomedical Research	6	cancer, cells, tumour, game, evolutionary, cell, molecular, theory, stage, mutations	17
Biomedical Research	7	energy, power, algorithm, proposed, green, game, certificate, nash, based, participants	13
Biomedical Research	8	quantum, classical, player, rational, game, setting, epr, secret, games, factorizable	5
Computer Science	0	game, network, defense, based, networks, proposed, attack, security, model, strategy	271
Computer Science	1	games, game, learning, agents, based, player, equilibrium, nash, algorithm, algorithms	196
Computer Science	2	energy, edge, game, computing, cloud, users, proposed, offloading, based, power	182

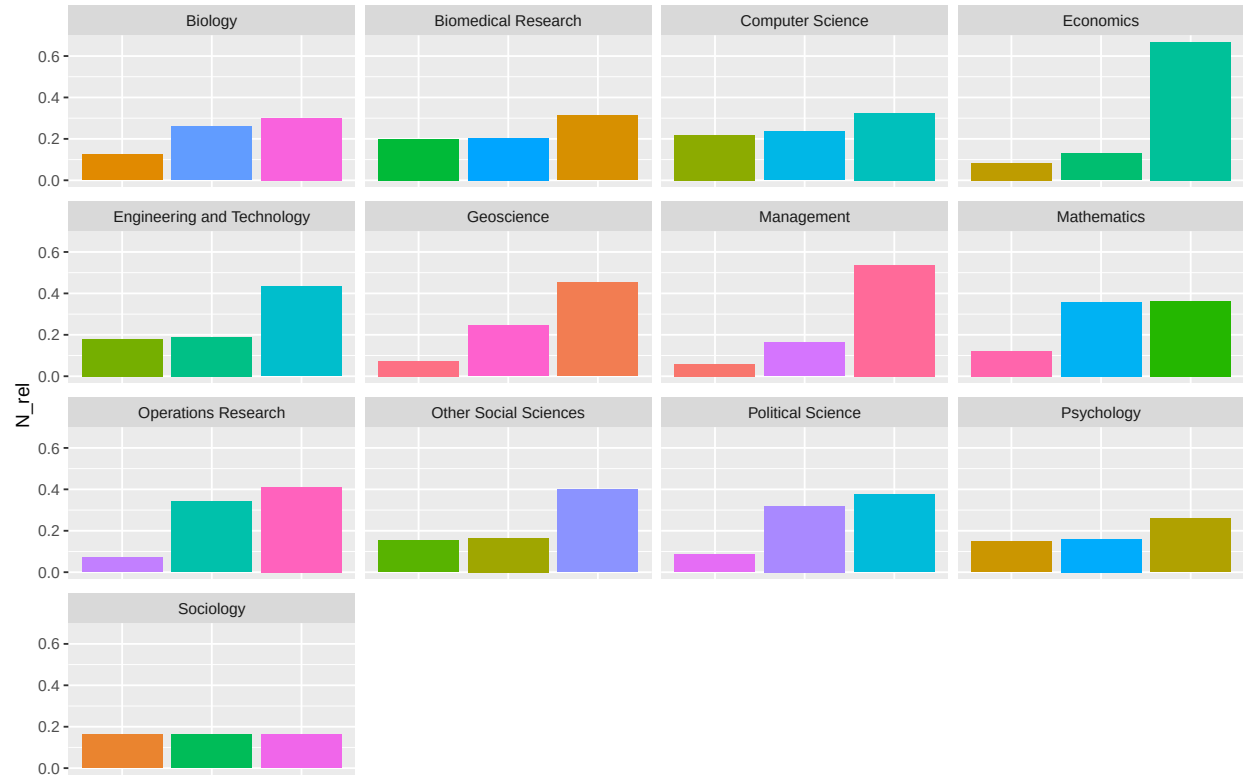
Computer Science	3	evolutionary, government, enterprises, game, innovation, model, supervision, public, strategy, evolution	74
Computer Science	4	fuzzy, control, game, cooperative, robust, proposed, theory, set, based, approach	50
Computer Science	5	supply, chain, product, pricing, retailer, manufacturer, advertising, carbon, paper, products	47
Computer Science	6	gan, deep, saliency, triple, pooling, player, learning, generative, classifier, supervised	5
Computer Science	7	manufacturing, cloud, production, service, platform, innovation, shop, evaluation, strategy, method	4
Computer Science	8	gate, noise, crosstalk, sizing, design, interconnect, discipline, optimization, designers, delay	3
Economics	0	game, games, equilibrium, players, repeated, theory, equilibria, model, strategies, payoffs	370
Economics	1	firms, firm, product, market, price, equilibrium, competition, model, dynamic, game	72
Economics	2	countries, international, game, trade, climate, environmental, global, power, gas, model	47
Economics	3	evolutionary, enterprises, government, game, sustainable, green, diffusion, model, development, chain	23
Economics	4	fisheries, resource, commons, gm, paper, externalities, fishery, costs, efficiency, harvest	14
Economics	5	matching, mechanism, teachers, sided, exposure, matches, preferences, stable, manipulable, problem	9
Economics	6	inflation, policy, bank, fiscal, central, monetary, sector, private, ramsey, targeting	9
Economics	7	corruption, tax, compliance, bribe, taxpayers, bureaucrat, extortion, compliant, honest, audit	7
Economics	8	morality, markets, free, norms, virtues, generalised, market, freedom, normative, societies	4
Engineering and Technology	0	game, network, networks, users, power, proposed, based, algorithm, problem, equilibrium	689
Engineering and Technology	1	game, games, control, nash, equilibrium, strategy, time, based, stochastic, agents	301
Engineering and Technology	2	energy, market, power, game, electricity, proposed, based, demand, model, trading	285
Engineering and Technology	3	game, cyber, attacks, security, attack, defense, mining, attacker, network, model	95
Engineering and Technology	4	government, carbon, evolutionary, game, emission, energy, model, construction, development, subsidy	82
Engineering and Technology	5	privacy, data, fl, incentive, mechanism, crowdsensing, learning, federated, model, users	64
Engineering and Technology	6	product, supply, chain, manufacturer, remanufacturing, advertising, retailer, products, retailers, supplier	45
Engineering and Technology	7	adversarial, learning, data, models, neural, machine, watermarking, attacks, game, image	17
Engineering and Technology	8	lt, gt, opinion, communities, social, opinions, community, networks, terrorist, nodes	16
Geoscience	0	carbon, government, evolutionary, game, green, enterprises, energy, environmental, model, strategy	155

Geoscience	1	supply, chain, carbon, low, retailer, products, manufacturer, blockchain, channel, remanufacturing	84
Geoscience	2	water, cooperative, allocation, river, resources, methods, decision, game, basin, theory	25
Geoscience	3	emergency, government, game, care, evolutionary, health, strategy, organizations, public, emergencies	21
Geoscience	4	innovation, knowledge, enterprises, collaborative, evolutionary, pharmaceutical, game, internet, platform, industry	21
Geoscience	5	climate, land, game, conservation, organizations, farmers, irrigation, change, foresight, lost	13
Geoscience	6	logistics, transport, alliance, border, cross, game, integration, delivery, commerce, service	10
Geoscience	7	safety, construction, supervision, miners, railway, government, workers, evolutionary, intercity, new	9
Geoscience	8	parking, lane, changing, traffic, metro, car, vehicle, cavs, model, vehicles	5
Health	0	elderly, platforms, care, service, evolutionary, patients, game, information, medical, mhealth	9
Health	1	quality, government, behavior, governance, enterprises, game, evolutionary, safety, media, environmental	13
Humanities	0	social, capital, individuals, putnam, theory, game, al, et, action, norms	9
Humanities	1	game, theoretic, word, theory, evolutionary, model, use, based, optimality, pragmatics	11
Management	0	supply, model, channel, market, price, product, chain, manufacturer, platform, game	308
Management	1	mml, game, games, team, theory, strategic, decision, cooperative, knowledge, value	94
Management	2	blockchain, evolutionary, game, platform, technology, enterprises, financial, model, financing, platforms	34
Management	3	evolutionary, government, green, carbon, game, enterprises, innovation, public, development, technology	33
Management	4	innovation, strategic, game, options, evolution, evolutionary, competitive, theory, firm, patent	29
Management	5	security, defender, attacker, hacker, source, attack, game, target, open, defense	27
Management	6	auctions, auction, bargaining, advertisers, price, bidding, seller, contests, search, crowdsourcing	27
Management	7	stochastic, markov, risk, equilibrium, state, dynamic, games, model, time, problem	18
Management	8	dea, efficiency, dmus, hospitals, internal, level, game, generalised, upper, envelopment	4
Mathematics	0	evolutionary, game, government, model, enterprises, strategies, strategy, innovation, cooperation, results	230
Mathematics	1	games, game, mml, equilibrium, players, stochastic, player, equilibria, nash, value	227
Mathematics	2	supply, chain, manufacturer, retailer, green, channel, pricing, product, price, products	77

Mathematics	3	game, attack, attacker, defense, network, modeling, adversarial, attacks, model, strategy	26
Mathematics	4	learning, algorithm, distributed, proposed, based, reinforcement, nash, control, adaptive, time	25
Mathematics	5	reinsurance, risk, investment, optimal, variance, time, consistent, asset, insurance, mean	22
Mathematics	6	stability, bifurcation, delays, dynamics, delay, game, imitative, hopf, time, nash	12
Mathematics	7	blockchain, agricultural, organizations, technology, governance, traceability, based, smart, products, supply	11
Mathematics	8	evacuation, emergency, model, cfs, ptn, exit, evacuees, congestion, based, fldr	4
Medicine	0	vaccination, control, disease, population, attack, security, vaccine, coverage, game, optimal	10
Medicine	1	tumor, cancer, cells, growth, cell, virulence, cycles, game, tumors, genes	8
Medicine	2	game, knowledge, patient, cooperation, medical, strategy, theory, care, management, physicians	9
Medicine	3	game, games, social, neural, choice, network, human, model, learning, behavior	24
Operations Research	0	supply, chain, manufacturer, retailer, channel, model, product, game, pricing, price	77
Operations Research	1	game, games, players, nash, mml, network, equilibria, equilibrium, algorithm, networks	64
Operations Research	2	lane, vehicles, traffic, vehicle, change, based, route, game, fairness, driver	13
Operations Research	3	reinsurance, investment, equilibrium, variance, pension, jump, asset, optimal, risky, risk	10
Operations Research	4	search, attacker, defender, game, warnings, defense, attack, terrorism, targets, theory	6
Operations Research	5	bus, passengers, passenger, competition, operators, road, traffic, equilibrium, service, railway	6
Operations Research	6	method, stochastic, proper, path, games, homotopy, compute, ipm, differentiable, equilibrium	4
Operations Research	7	pollution, dea, dmus, decision, medical, jme, proposed, efficiency, units, treatment	4
Operations Research	8	management, operation, situation, strict, speed, safety, ptps, ctps, aviation, civil	3
Other Professional Fields	0	game, shipping, competition, capacity, container, sharing, strategy, supply, ports, demand	19
Other Professional Fields	1	outsourcing, theory, relational, motivations, information, decision, core, case, strategic, firm	5
Other Professional Fields	2	product, market, model, content, free, firms, service, online, quality, firm	14
Other Professional Fields	3	game, play, video, group, groups, research, students, theory, experiment, games	7
Other Professional Fields	4	traffic, approach, problem, toll, game, assignment, school, cavs, operators, vehicle	8
Other Social Sciences	0	infrastructure, interdependent, network, restoration, decentralized, recovery, networks, decision, decisions, controllers	5

Other Social Sciences	1	defender, security, attacker, cyber, risk, allocation, game, defense, attacks, model	14
Other Social Sciences	2	gt, lt, li, game, social, learning, theory, evolutionary, evolution, strategy	34
Other Social Sciences	3	economic, theory, political, institutions, international, institutional, neoclassical, economics, new, article	7
Other Social Sciences	4	facilities, nimby, community, dilemma, institutions, natural, prisoner, tourism, resource, institutional	5
Other Social Sciences	5	manufacturing, cloud, equilibrium, local, multiple, platforms, competition, game, platform, location	7
Other Social Sciences	6	environmental, carbon, enterprises, evolutionary, game, government, public, regulation, pollution, technology	13
Physics	0	game, evolutionary, cooperation, games, strategy, dynamics, strategies, model, population, payoff	251
Physics	1	energy, game, power, users, network, wireless, nodes, sensor, proposed, algorithm	58
Physics	2	quantum, classical, theory, games, game, players, entanglement, strategies, eve, alice	28
Physics	3	vaccination, epidemic, strategy, model, individuals, disease, conformity, strain, cost, vaccine	11
Physics	4	infrastructure, attacker, attack, information, defender, networks, network, attacks, strategies, performance	10
Physics	5	chain, supply, channel, manufacturer, retailer, agency, product, profit, service, dual	10
Physics	6	radar, allocation, target, imaging, task, power, game, problem, time, isar	9
Physics	7	learning, policy, adaptive, optimal, tracking, control, based, graphical, poker, integral	8
Physics	8	mean, field, equations, algorithm, problem, discrete, parabolic, kolmogorov, decentralised, control	5
Political Science	0	game, political, theory, policy, theoretic, party, model, preferences, equilibrium, legislative	87
Political Science	1	international, game, cooperation, model, bargaining, conflict, war, theory, states, theoretic	74
Political Science	2	policy, gas, eu, pg, model, european, making, states, decision, russia	20
Political Science	3	terrorism, terrorist, note, 2010, al, security, attacks, 2009, intelligence, december	12
Political Science	4	eu, commission, european, council, intergovernmental, union, activity, policy, member, parliament	11
Political Science	5	rebel, groups, violence, indigenous, civil, rebels, government, wars, civilian, fighting	9
Political Science	6	sanctions, sender, economic, target, coercion, senders, secondary, model, sanction, enforcement	9
Political Science	7	linguistic, fuzzy, logic, game, false, logical, piracy, hobbes, rational, true	5
Political Science	8	repression, media, rights, protest, human, sexual, preemptive, government, oversight, abuses	5

Psychology	0	decision, strategic, information, game, theory, rationality, human, making, interaction, players	21
Psychology	1	government, evolutionary, game, governments, pattern, safety, theory, supervision, china, local	13
Psychology	2	cooperation, prisoner, dilemma, punishment, repeated, games, behavior, strategies, experiments, reasoning	12
Psychology	3	size, fish, cleaner, cleaners, predator, individuals, prey, client, body, feeding	9
Psychology	4	evolution, population, dynamics, evolutionary, theory, behavior, learning, intention, game, language	7
Psychology	5	game, social, preferences, projection, prosocial, krueger, games, theory, moral, initial	7
Psychology	6	coalition, conflict, coalitions, nuclear, negotiation, teams, bargaining, model, cts, power	6
Psychology	7	conditional, successful, evolutionary, spiteful, strategies, spite, machiavellianism, repeated, different, altruism	3
Psychology	8	weaker, competition, stronger, dominance, winner, conflicts, relative, chance, competitor, compete	3
Sociology	0	crime, incentives, inspection, control, learning, cause, punishment, stronger, rationality, prisoner	6
Sociology	1	exchange, network, networks, theories, power, theory, core, effectiveness, models, solution	7
Sociology	2	power, games, game, structures, theory, coalitions, pareto, high, low, dilemma	7
Sociology	3	commitment, actors, average, strategy, information, interactions, actor, weak, performance, note	7
Sociology	4	game, trust, evolution, ultimatum, partner, fairness, model, institutions, self, command	6
Sociology	5	queen, king, caesar, dupes, equilibria, rational, perfectly, skills, narrative, frustration	4
Sociology	6	gt, interaction, social, strategic, players, depends, violations, practices, origins, transitivity	3
Sociology	7	theory, science, use, game, research, lobster, maine, limitations, able, stages	3

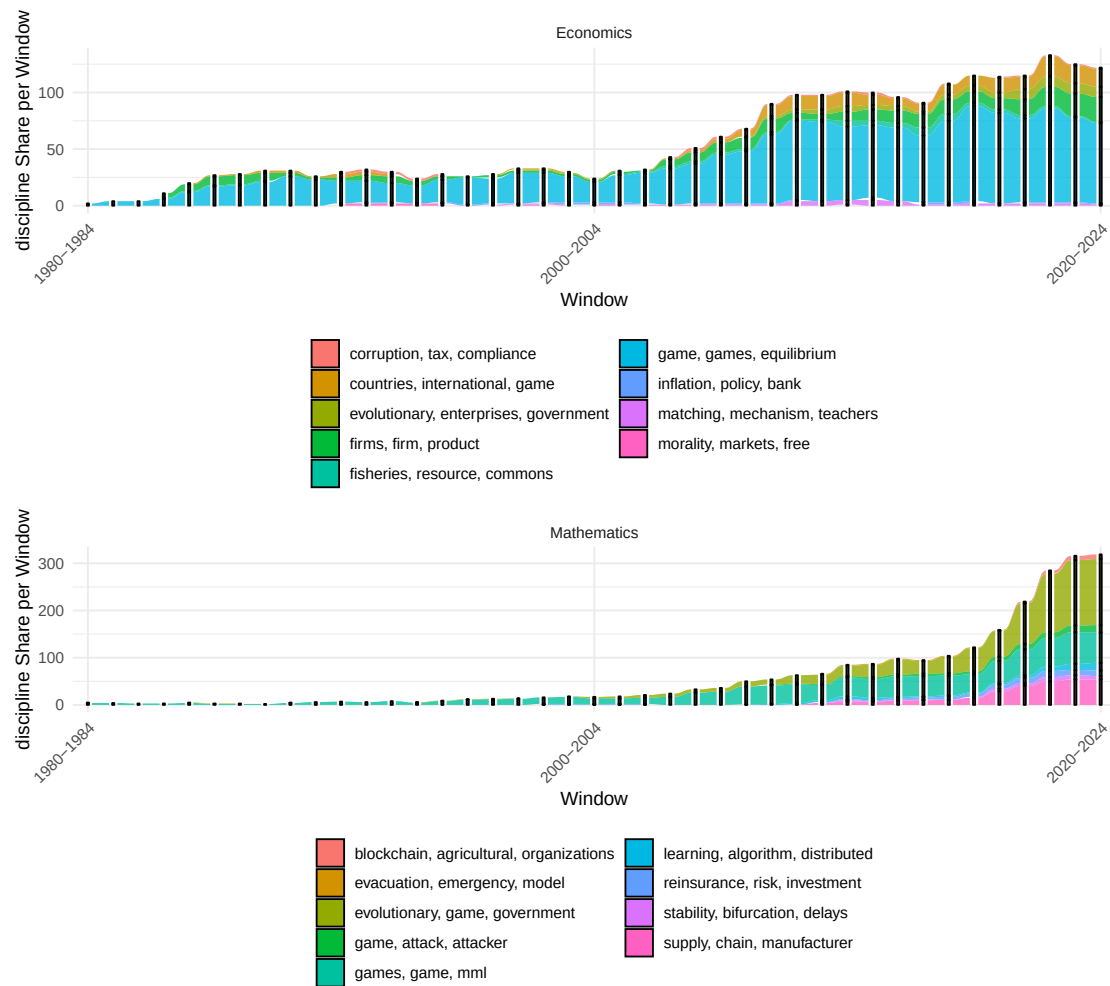


Keywords

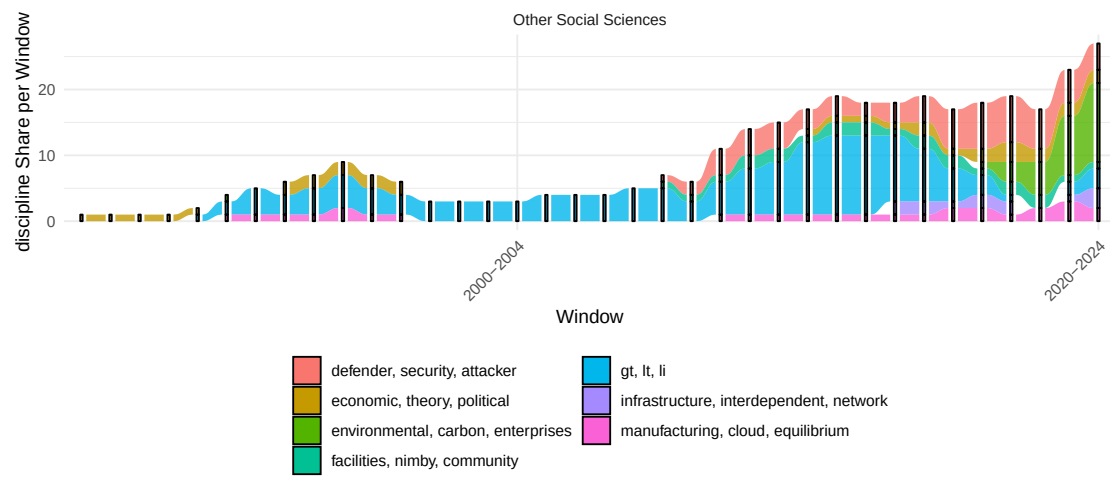
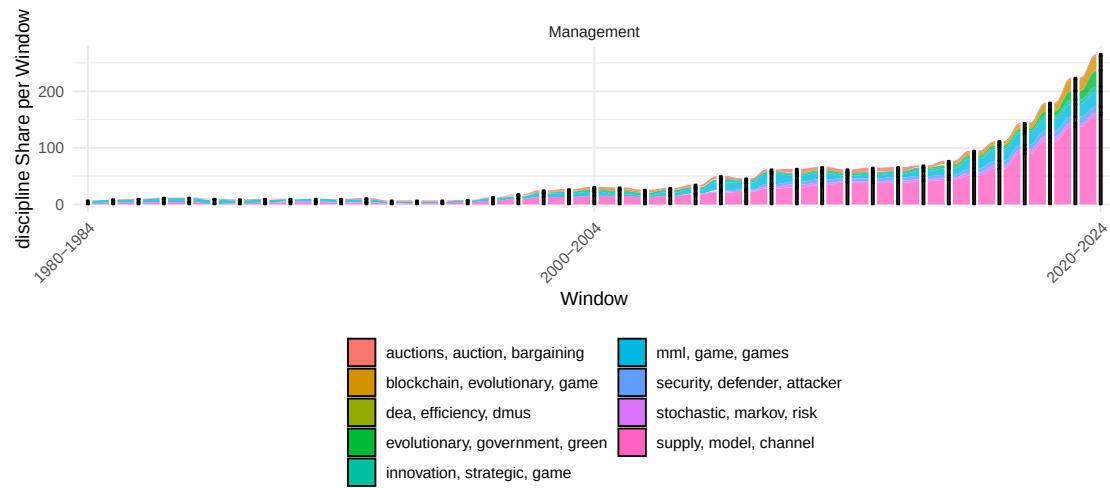
blockchain, evolutionary, game, platform, technology, enterprises, financial, model, financing, platforms	game, network, networks, users, power, proposed, based, algorithm, problem, equilibrium
carbon, government, evolutionary, game, green, enterprises, energy, environmental, model, strategy	game, political, theory, policy, theoretic, party, model, preferences, equilibrium, legislative
commitment, actors, average, strategy, information, interactions, actor, weak, performance, note	games, game, learning, agents, based, player, equilibrium, nash, algorithm, algorithms
cooperation, game, social, games, trust, selection, evolutionary, defectors, strategies, players	games, game, mml, equilibrium, players, stochastic, player, equilibria, nash, value
cooperation, game, social, group, evolution, theory, individuals, model, helping, evolutionary	government, evolutionary, game, governments, pattern, safety, theory, supervision, china, local
cooperation, prisoner, dilemma, punishment, repeated, games, behavior, strategies, experiments, reasoning	government, game, evolutionary, strategy, carbon, chain, enterprises, model, supply, low
countries, international, game, trade, climate, environmental, global, power, gas, model	government, marine, game, management, enterprises, development, study, governance, carbon, food
decision, strategic, information, game, theory, rationality, human, making, interaction, players	gt, lt, ll, game, social, learning, theory, evolutionary, evolution, strategy
defender, security, attacker, cyber, risk, allocation, game, defense, attacks, model	international, game, cooperation, model, bargaining, conflict, war, theory, states, theoretic
energy, edge, game, computing, cloud, users, proposed, offloading, based, power	lane, vehicles, traffic, vehicle, change, based, route, game, fairness, driver
energy, market, power, game, electricity, proposed, based, demand, model, trading	mml, game, games, team, theory, strategic, decision, cooperative, knowledge, value
environmental, carbon, enterprises, evolutionary, game, government, public, regulation, pollution, technology	policy, gas, eu, pg, model, european, making, states, decision, russia
evolutionary, game, government, model, enterprises, strategies, strategy, innovation, cooperation, results	power, games, game, structures, theory, coalitions, pareto, high, low, dilemma
evolutionary, population, game, dynamics, host, model, species, strategies, selection, strategy	prey, males, male, food, mating, female, game, model, females, behavior
exchange, network, networks, theories, power, theory, core, effectiveness, models, solution	supply, chain, carbon, low, retailer, products, manufacturer, blockchain, channel, remanufacturing
firms, firm, product, market, price, equilibrium, competition, model, dynamic, game	supply, chain, manufacturer, retailer, channel, model, product, game, pricing, price
game, games, control, nash, equilibrium, strategy, time, based, stochastic, agents	supply, chain, manufacturer, retailer, green, channel, pricing, product, price, products
game, games, equilibrium, players, repeated, theory, equilibria, model, strategies, payoffs	supply, model, channel, market, price, product, chain, manufacturer, platform, game
game, games, players, nash, mml, network, equilibria, equilibrium, algorithm, networks	water, cooperative, allocation, river, resources, methods, decision, game, basin, theory
game, network, defense, based, networks, proposed, attack, security, model, strategy	

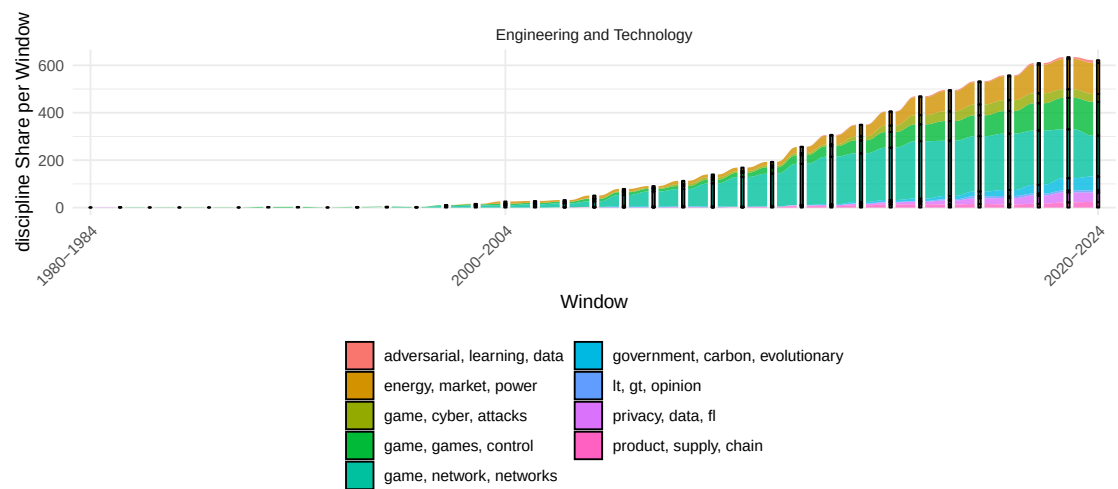
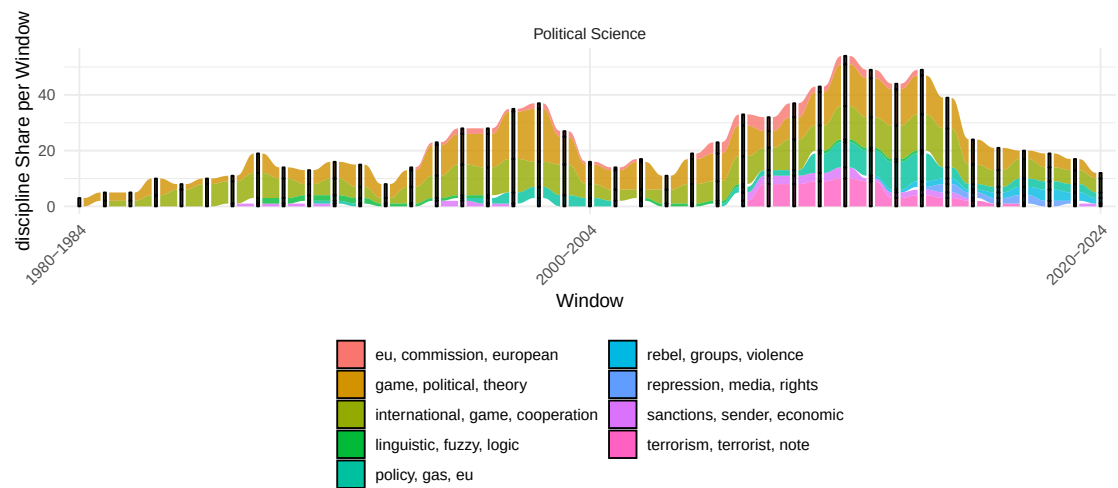
5.2 Evolution of Topics over Time

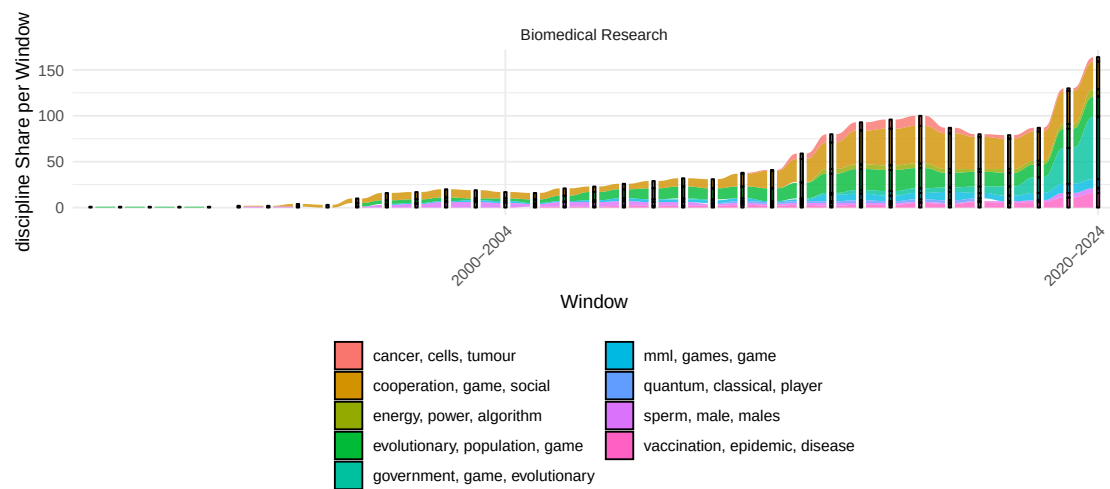
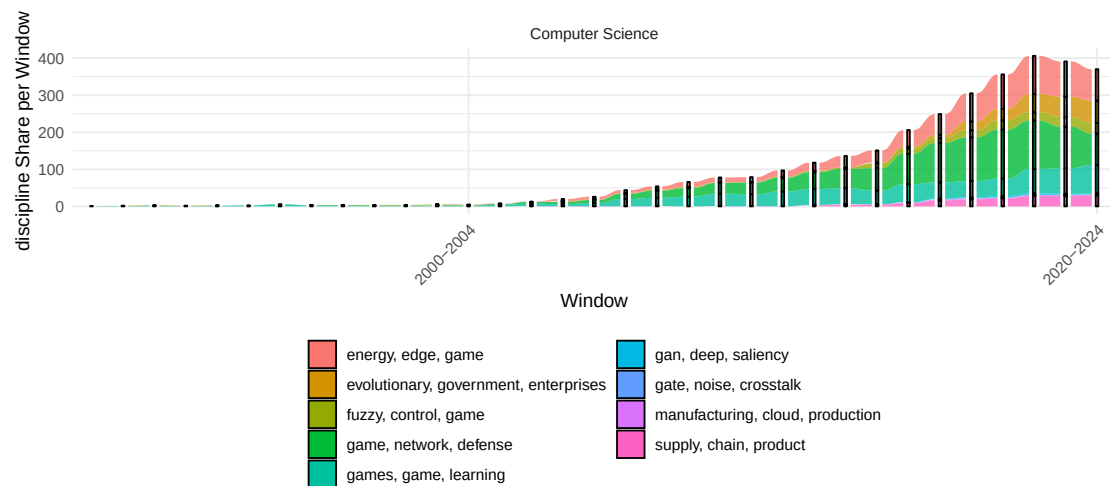
5.2.1 Based on N

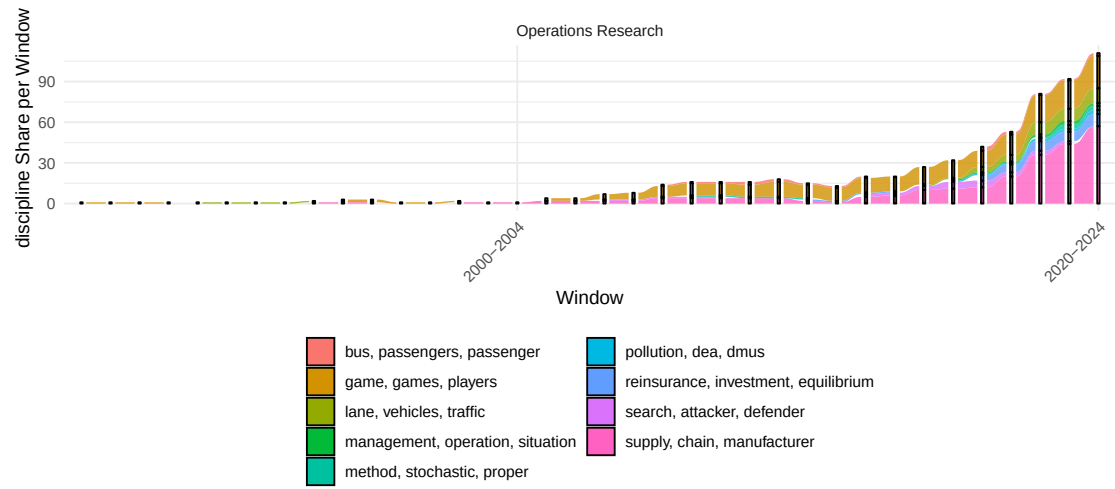
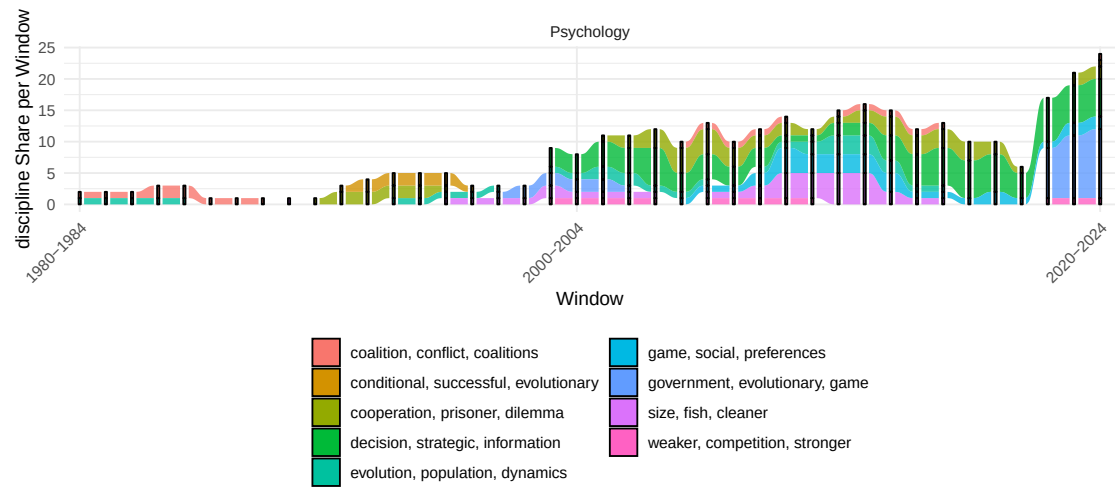


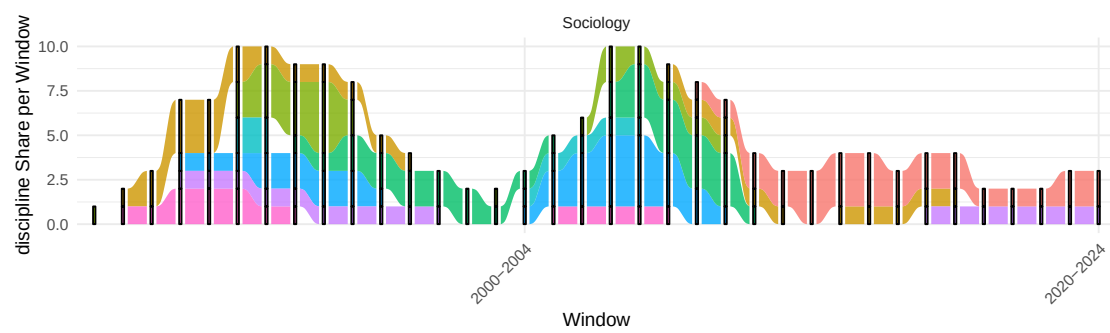
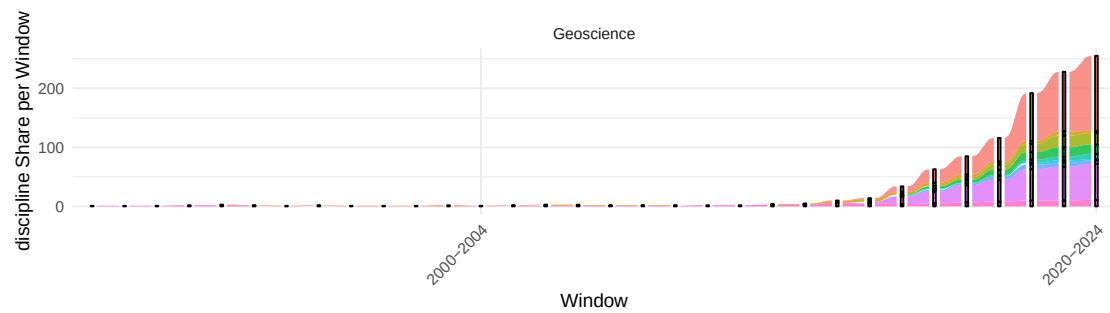
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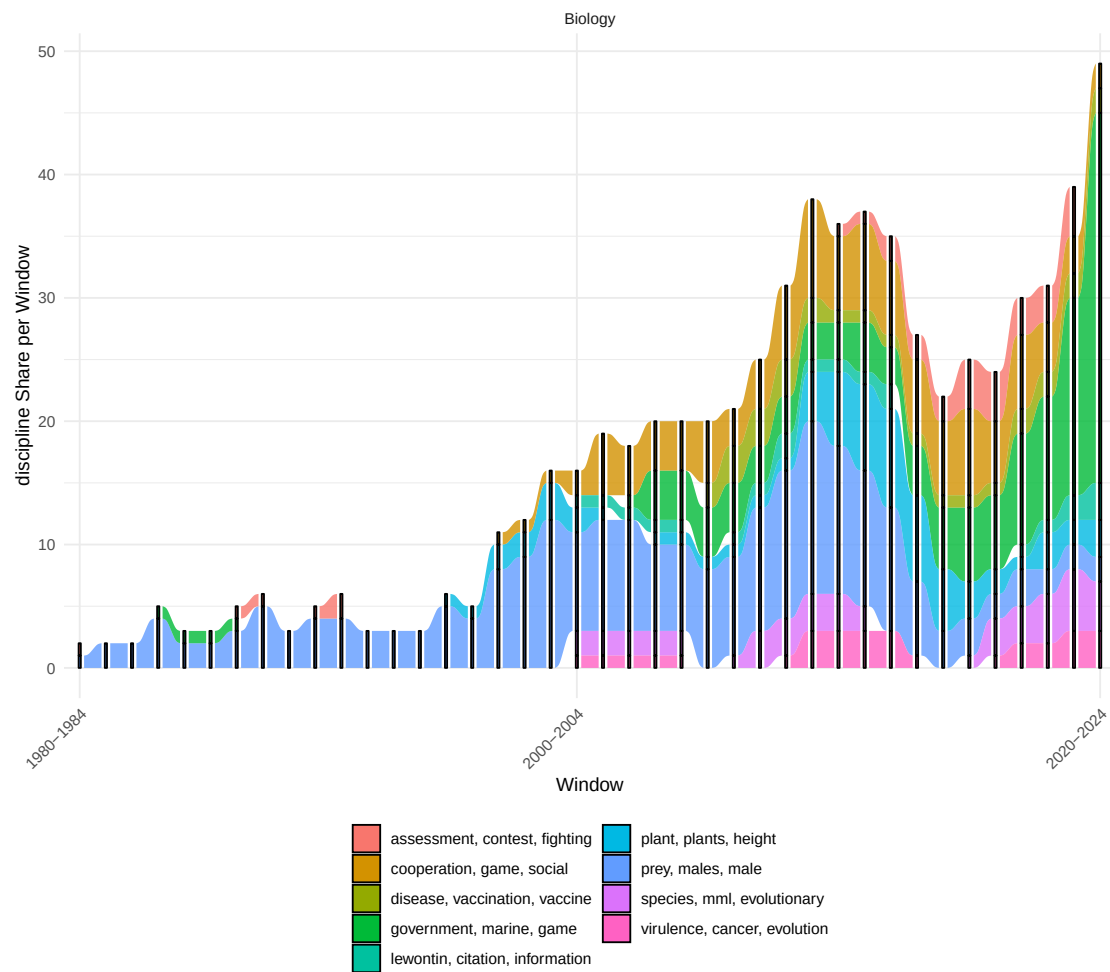




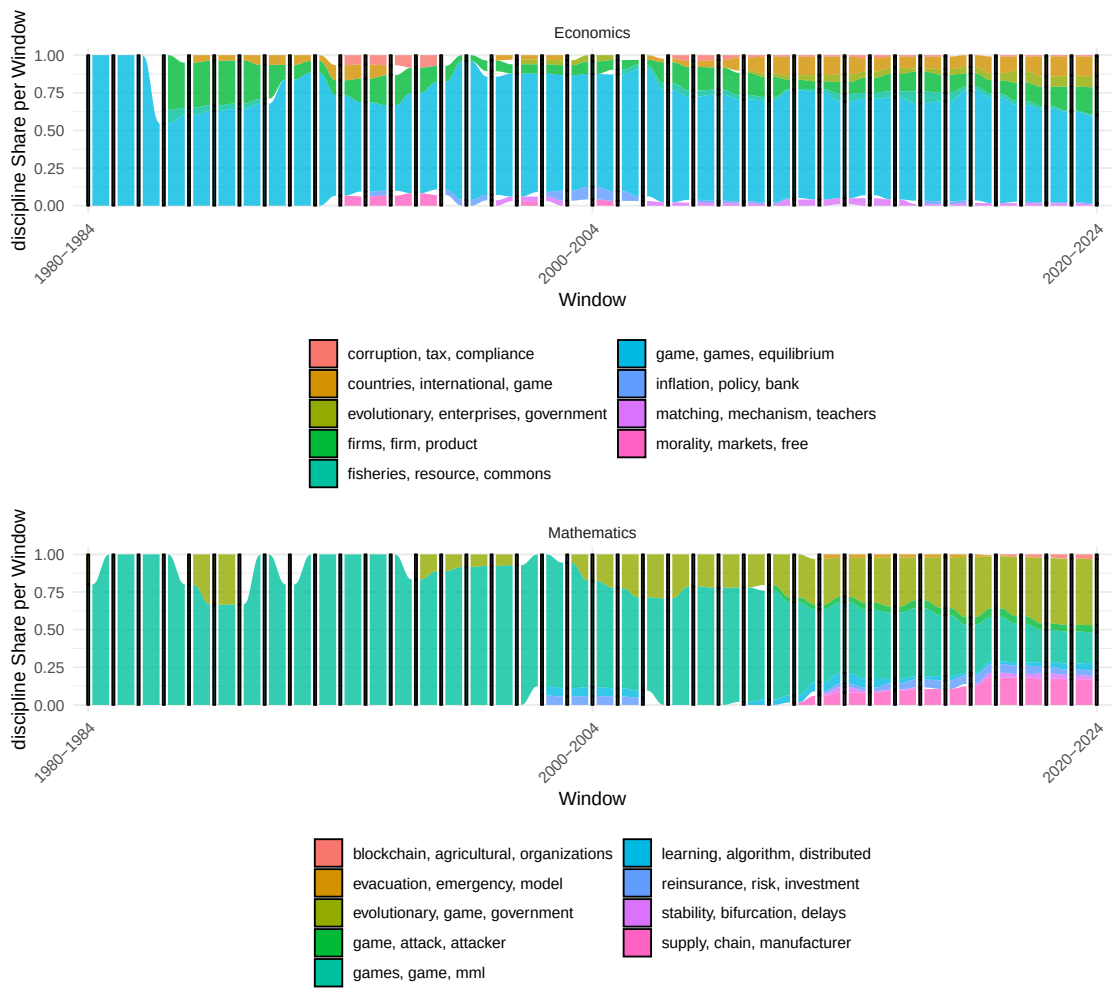








5.2.2 Based on N^{rel}



[1] “
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